

PBSAI Control Mapping to NIST SP 800-53 Revision 5

Enterprise Defensive Cybersecurity Control Crosswalk for AI-Enabled Security Operations

Abstract

The Practitioner's Blueprint for Secure AI (PBSAI) Defensive Cybersecurity Multi-Agent Ecosystem is a governance-centered, security-oriented reference architecture for securing enterprise AI estates through coordinated, policy-bound autonomous and semi-autonomous agents.

This document provides a comprehensive mapping between PBSAI controls and NIST SP 800-53 Revision 5 security and privacy controls.

PBSAI is organized across twelve security and governance domains, supported by global controls for context envelopes, output contracts, evidence generation, provenance, human-in-the-loop authorization, safety, rollback, and feedback. The PBSAI Control Catalog already defines PBSAI's control families and maps them to NIST SP 800-53 at both global and domain levels.

This mapping document is intended to provide a formal control crosswalk suitable for:

- enterprise security architecture,
 - governance and compliance analysis,
 - OLIR-oriented mapping development,
 - control inheritance analysis,
 - audit-readiness planning,
 - AI-enabled defensive cybersecurity system design,
 - and PBSAI implementation planning.
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1. Introduction

PBSAI provides a domain-partitioned operational realization model for enterprise cybersecurity governance in AI-enabled environments. It is designed to operate as a defensive cybersecurity overlay across CISO functions and can either govern existing tools or provide direct agentic functionality through controlled automation.

PBSAI's control model is organized around:

- global governance controls,
- domain-specific cybersecurity controls,
- evidence and auditability requirements,
- provenance and attestation requirements,
- human authorization requirements,
- output governance,
- feedback and continuous-improvement mechanisms,
- and operational domain controls spanning A through L.

The purpose of this document is to map PBSAI controls to NIST SP 800-53 Revision 5 controls and control enhancements in a structured, traceable, and implementation-oriented format.

2. Scope and Methodology

This document maps PBSAI controls to NIST SP 800-53 Revision 5 using the following interpretive logic:

- PBSAI controls are mapped at the individual control level.
- Each mapping identifies relevant NIST SP 800-53 controls or enhancements.
- Relationship semantics distinguish direct implementation from supporting or enhancing alignment.
- Rationale explains why the mapping is appropriate.
- The mapping is control-oriented, not agent-oriented.

PBSAI controls often map to multiple NIST controls because PBSAI controls are designed as operational mechanisms. For example, a PBSAI evidence control may support audit generation, audit content, non-repudiation, and continuous assessment simultaneously.

3. Relationship Semantics

Relationship Definition

equivalent	PBSAI materially implements the NIST control objective
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Relationship	Definition
strongly-supports	PBSAI provides major operational support for the NIST control objective
supports	PBSAI partially contributes to the NIST control objective
enhances	PBSAI adds execution semantics, automation, provenance, or evidence chaining beyond the base NIST control
constrains	PBSAI enforces an operational boundary or limitation associated with the NIST control objective

4. Mapping Rules

This document uses the following mapping rules:

1. **Primary implementation mappings** are marked equivalent where the PBSAI control directly implements the NIST control intent.
 2. **Operational support mappings** are marked strongly-supports or supports where PBSAI supplies evidence, telemetry, context, provenance, policy enforcement, or decision support.
 3. **Enhanced governance mappings** are marked enhances where PBSAI extends the NIST control intent through automation, structured evidence, provenance, deterministic policy binding, or AI-enabled security operations.
 4. **Boundary and enforcement mappings** are marked constrains where PBSAI limits execution scope, access scope, data movement, system behavior, or response authority.
 5. This mapping does not claim that PBSAI alone satisfies all organizational, procedural, legal, or administrative obligations associated with NIST SP 800-53. It identifies how PBSAI controls implement, support, or enhance NIST control objectives.
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5. Global PBSAI Controls → NIST SP 800-53 Rev 5

The global PBSAI controls define reusable execution and governance semantics across the entire ecosystem, including context and policy binding, Output Contract governance, evidence generation, provenance, human authorization, safety, rollback, and feedback.

PBSAI-GEN-MCP — Context and Policy Binding

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-MCP-001 Context Envelope Requirement	AC-3 Access Enforcement	strongly-supports	Provides structured context required for authorization and enforcement decisions
PBSAI-GEN-MCP-001 Context Envelope Requirement	AC-16 Security and Privacy Attributes	equivalent	Encapsulates identity, classification, policy, and constraint attributes for downstream control evaluation
PBSAI-GEN-MCP-001 Context Envelope Requirement	AU-3 Content of Audit Records	supports	Supplies contextual fields required for meaningful audit reconstruction
PBSAI-GEN-MCP-002 Context Completeness Enforcement	SI-10 Information Input Validation	equivalent	Validates completeness and correctness of governance input context
PBSAI-GEN-MCP-002 Context Completeness Enforcement	AC-3 Access Enforcement	strongly-supports	Prevents authorization decisions based on incomplete context
PBSAI-GEN-MCP-002 Context Completeness Enforcement	IA-5 Authenticator Management	supports	Supports validation of identity/authentication-related context before decisions
PBSAI-GEN-MCP-003 Context Propagation	AU-12 Audit Record Generation	strongly-supports	Maintains context continuity across domain interactions and audit records

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-MCP-003 Context Propagation	AC-16 Security and Privacy Attributes	strongly-supports	Propagates security and privacy attributes across workflows
PBSAI-GEN-MCP-003 Context Propagation	SI-4 System Monitoring	supports	Enables context-rich monitoring and correlation

PBSAI-GEN-OC — Output Contract Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-OC-001 Output Contract Requirement	SI-10 Information Input Validation	equivalent	Requires machine-readable output schemas suitable for validation before acceptance or execution
PBSAI-GEN-OC-001 Output Contract Requirement	SA-10 Developer Configuration Management	supports	Supports controlled interface and artifact definitions for automated systems
PBSAI-GEN-OC-001 Output Contract Requirement	CM-3 Configuration Change Control	supports	Ensures outputs that may drive changes are structured and governable
PBSAI-GEN-OC-002 Output Validation	SI-10 Information Input Validation	equivalent	Validates outputs before downstream use or execution
PBSAI-GEN-OC-002 Output Validation	AU-12 Audit Record Generation	supports	Produces validation events for auditability
PBSAI-GEN-OC-003 Output Rejection and Escalation	IR-4 Incident Handling	supports	Routes invalid or unsafe outputs to review or response workflows

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-OC-003 Output Rejection and Escalation	AC-3 Access Enforcement	supports	Prevents invalid outputs from becoming authorized actions
PBSAI-GEN-OC-003 Output Rejection and Escalation	AU-12 Audit Record Generation	supports	Records rejection and escalation outcomes

PBSAI-GEN-EVID — Evidence Generation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-EVID-001 Evidence Generation Requirement	AU-2 Event Logging	equivalent	Requires control actions to generate security-relevant evidence events
PBSAI-GEN-EVID-001 Evidence Generation Requirement	AU-3 Content of Audit Records	strongly-supports	Links inputs, decisions, policy references, actors, and outcomes
PBSAI-GEN-EVID-001 Evidence Generation Requirement	AU-12 Audit Record Generation	equivalent	Requires generation of structured evidence artifacts
PBSAI-GEN-EVID-002 Evidence Integrity	AU-9 Protection of Audit Information	equivalent	Protects evidence from unauthorized modification
PBSAI-GEN-EVID-002 Evidence Integrity	AU-10 Non-Repudiation	strongly-supports	Preserves evidence trustworthiness and attribution

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-EVID-002 Evidence Integrity	SI-7 Software, Firmware, and Information Integrity	supports	Supports integrity validation for evidence-bearing artifacts
PBSAI-GEN-EVID-003 Evidence Accessibility	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Makes evidence usable for audit, review, investigation, and analysis
PBSAI-GEN-EVID-003 Evidence Accessibility	CA-2 Control Assessments	supports	Supports evidence-based control assessment
PBSAI-GEN-EVID-003 Evidence Accessibility	CA-7 Continuous Monitoring	supports	Enables continuous use of evidence for compliance and control-health evaluation

PBSAI-GEN-PROV — Provenance and Attestation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-PROV-001 Provenance Attribution	AU-10 Non-Repudiation	equivalent	Attributes outputs to producing identities or systems
PBSAI-GEN-PROV-001 Provenance Attribution	IA-2 Identification and Authentication	supports	Links producer identity to governed output generation
PBSAI-GEN-PROV-001 Provenance Attribution	SR-11 Component Authenticity	strongly-supports	Supports authenticity of produced artifacts and outputs
PBSAI-GEN-PROV-002 Cryptographic Provenance	SI-7 Software, Firmware, and Information Integrity	equivalent	Uses cryptographic signatures to protect integrity and authenticity

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-PROV-002 Cryptographic Provenance	SC-12 Cryptographic Key Establishment and Management	supports	Relies on secure key management for provenance signatures
PBSAI-GEN-PROV-002 Cryptographic Provenance	AU-10 Non-Repudiation	strongly-supports	Ensures signed outputs can be attributed and verified
PBSAI-GEN-PROV-003 Provenance Validation	SR-4 Provenance	equivalent	Validates provenance before accepting outputs or artifacts
PBSAI-GEN-PROV-003 Provenance Validation	SR-11 Component Authenticity	strongly-supports	Rejects unauthenticated or unverifiable components/outputs
PBSAI-GEN-PROV-003 Provenance Validation	SI-10 Information Input Validation	supports	Treats missing or invalid provenance as invalid input

PBSAI-GEN-HITL — Human Authorization

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-HITL-001 Human Approval Requirement	AC-3 Access Enforcement	strongly-supports	Requires approval before execution of high-impact actions
PBSAI-GEN-HITL-001 Human Approval Requirement	AC-5 Separation of Duties	supports	Enables separation of automated proposal and human authorization
PBSAI-GEN-HITL-001 Human Approval Requirement	IR-4 Incident Handling	supports	Supports supervised response and escalation workflows

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-HITL-002 Approval Attribution	AU-10 Non-Repudiation	equivalent	Binds approvals to authenticated individuals
PBSAI-GEN-HITL-002 Approval Attribution	AU-3 Content of Audit Records	strongly-supports	Records approver identity, timestamp, and decision context
PBSAI-GEN-HITL-003 Multi-Party Authorization	AC-5 Separation of Duties	equivalent	Requires multiple approvers for high-risk actions
PBSAI-GEN-HITL-003 Multi-Party Authorization	AC-6 Least Privilege	supports	Restricts elevated authority to approved quorum conditions
PBSAI-GEN-HITL-003 Multi-Party Authorization	AU-12 Audit Record Generation	supports	Records multi-party authorization events

PBSAI-GEN-SAFE — Safety, Rollback, and Kill Switch

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-SAFE-001 Safe Execution Constraint	SI-13 Predictable Failure Prevention	strongly-supports	Requires safe operational behavior under failure or uncertainty
PBSAI-GEN-SAFE-001 Safe Execution Constraint	SC-24 Fail in Known State	equivalent	Ensures unsafe or uncertain states resolve to controlled behavior
PBSAI-GEN-SAFE-002 Rollback Requirement	CP-10 System Recovery and Reconstitution	strongly-supports	Enables restoration after unsafe or failed operational action

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-SAFE-002 Rollback Requirement	CM-3 Configuration Change Control	supports	Supports reversal of controlled changes
PBSAI-GEN-SAFE-003 Kill Switch / Emergency Stop	IR-4 Incident Handling	strongly-supports	Enables emergency containment or interruption of unsafe behavior
PBSAI-GEN-SAFE-003 Kill Switch / Emergency Stop	SI-13 Predictable Failure Prevention	supports	Prevents continued unsafe system behavior
PBSAI-GEN-SAFE-003 Kill Switch / Emergency Stop	SC-7 Boundary Protection	supports	Can isolate or constrain system boundaries during emergency conditions

PBSAI-GEN-FEED — Feedback and Continuous Improvement

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-FEED-001 Feedback Capture	CA-7 Continuous Monitoring	strongly-supports	Captures operational findings for continuous control evaluation
PBSAI-GEN-FEED-001 Feedback Capture	PM-14 Testing, Training, and Monitoring	supports	Supports organizational learning from operational evidence
PBSAI-GEN-FEED-002 Control Improvement	PM-14 Testing, Training, and Monitoring	strongly-supports	Uses outcomes to improve controls and governance processes
PBSAI-GEN-FEED-002 Control Improvement	RA-7 Risk Response	supports	Uses findings to improve mitigation and response actions

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-GEN-FEED-003 Cross-Domain Feedback Integration	CA-7 Continuous Monitoring	strongly-supports	Feeds findings across monitoring, response, governance, and risk domains
PBSAI-GEN-FEED-003 Cross-Domain Feedback Integration	SI-4 System Monitoring	supports	Improves monitoring effectiveness through feedback loops
PBSAI-GEN-FEED-003 Cross-Domain Feedback Integration	IR-4 Incident Handling	supports	Improves incident response based on prior outcomes

6. Domain A — Governance, Risk, and Compliance → NIST SP 800-53 Rev 5

Domain A defines PBSAI's governance, risk, compliance, monitoring, and supplier-risk functions. The control catalog identifies Domain A as covering policy authority, risk computation, compliance validation, monitoring and analytics, and supplier risk oversight.

A-POL — Policy Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-POL-001 Policy Definition and Maintenance	PL-1 Policy and Procedures	equivalent	Requires formal governance policies and control objectives
PBSAI-A-POL-001 Policy Definition and Maintenance	PM-1 Information Security Program Plan	supports	Establishes program-level governance direction
PBSAI-A-POL-001 Policy Definition and Maintenance	PM-9 Risk Management Strategy	supports	Connects governance policy to risk-management objectives

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-POL-002 Policy Versioning and Traceability	PL-2 System Security and Privacy Plans	supports	Links policy versions to system governance context
PBSAI-A-POL-002 Policy Versioning and Traceability	CM-3 Configuration Change Control	strongly-supports	Treats policy changes as governed change events
PBSAI-A-POL-002 Policy Versioning and Traceability	AU-12 Audit Record Generation	supports	Records policy-version usage in decisions
PBSAI-A-POL-003 Policy Distribution and Availability	PL-2 System Security and Privacy Plans	supports	Ensures policies are available to governed systems
PBSAI-A-POL-003 Policy Distribution and Availability	AC-3 Access Enforcement	supports	Provides policies needed for enforcement decisions
PBSAI-A-POL-003 Policy Distribution and Availability	CA-7 Continuous Monitoring	supports	Enables continuous monitoring against current policy
PBSAI-A-POL-004 Policy Change Control	CM-3 Configuration Change Control	equivalent	Governs changes to policies as controlled modifications
PBSAI-A-POL-004 Policy Change Control	CM-5 Access Restrictions for Change	supports	Restricts policy modification authority
PBSAI-A-POL-004 Policy Change Control	AU-12 Audit Record Generation	supports	Records policy change approvals and history
PBSAI-A-POL-005 Policy Conflict Resolution	PL-2 System Security and Privacy Plans	supports	Resolves conflicts affecting system security planning

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-POL-005 Policy Conflict Resolution	RA-3 Risk Assessment	supports	Evaluates risk from conflicting policies
PBSAI-A-POL-005 Policy Conflict Resolution	SI-10 Information Input Validation	enhances	Prevents inconsistent policy logic from being accepted into evaluation
PBSAI-A-POL-006 Policy Completeness and Coverage	CA-2 Control Assessments	strongly-supports	Identifies whether controls and governed actions are adequately covered
PBSAI-A-POL-006 Policy Completeness and Coverage	CA-7 Continuous Monitoring	supports	Supports ongoing coverage evaluation
PBSAI-A-POL-006 Policy Completeness and Coverage	PL-2 System Security and Privacy Plans	supports	Ensures policy coverage aligns with system security planning
PBSAI-A-POL-007 Policy Enforcement Readiness	SA-8 Security and Privacy Engineering Principles	strongly-supports	Ensures policies are implementable and testable
PBSAI-A-POL-007 Policy Enforcement Readiness	SI-10 Information Input Validation	supports	Requires structured, unambiguous policy logic
PBSAI-A-POL-007 Policy Enforcement Readiness	AC-3 Access Enforcement	supports	Makes policies usable by enforcement systems

A-RSK — Risk Management

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-RSK-001 Risk Assessment Requirement	RA-3 Risk Assessment	equivalent	Requires structured risk assessments for governed assets, actions, and states
PBSAI-A-RSK-001 Risk Assessment Requirement	RA-1 Policy and Procedures	supports	Establishes procedural basis for risk assessment
PBSAI-A-RSK-002 Risk Scoring and Quantification	RA-3 Risk Assessment	strongly-supports	Produces quantifiable risk scores and contributing factors
PBSAI-A-RSK-002 Risk Scoring and Quantification	RA-5 Vulnerability Monitoring and Scanning	supports	Supports vulnerability-related risk quantification
PBSAI-A-RSK-002 Risk Scoring and Quantification	PM-9 Risk Management Strategy	supports	Links quantified risk to strategy
PBSAI-A-RSK-003 Risk Context Integration	RA-3 Risk Assessment	strongly-supports	Incorporates asset, identity, threat, and environmental context
PBSAI-A-RSK-003 Risk Context Integration	AC-16 Security and Privacy Attributes	supports	Uses contextual attributes in risk evaluation
PBSAI-A-RSK-003 Risk Context Integration	CM-8 System Component Inventory	supports	Uses asset inventory as risk context
PBSAI-A-RSK-004 Continuous Risk Evaluation	RA-3 Risk Assessment	enhances	Continuously updates risk based on changing system and threat conditions
PBSAI-A-RSK-004 Continuous Risk Evaluation	RA-5 Vulnerability Monitoring and Scanning	supports	Incorporates vulnerability changes into risk evaluation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-RSK-004 Continuous Risk Evaluation	CA-7 Continuous Monitoring	strongly-supports	Provides event-driven risk updates
PBSAI-A-RSK-005 Risk-Based Decision Support	RA-7 Risk Response	equivalent	Links risk results to authorization, mitigation, remediation, or escalation
PBSAI-A-RSK-005 Risk-Based Decision Support	AC-3 Access Enforcement	supports	Supplies risk inputs for access/enforcement decisions
PBSAI-A-RSK-005 Risk-Based Decision Support	CA-7 Continuous Monitoring	supports	Supports ongoing risk-informed governance
PBSAI-A-RSK-006 Risk Prioritization	RA-7 Risk Response	strongly-supports	Prioritizes mitigation and remediation based on risk
PBSAI-A-RSK-006 Risk Prioritization	PM-9 Risk Management Strategy	supports	Aligns prioritization with risk strategy
PBSAI-A-RSK-006 Risk Prioritization	PM-11 Mission and Business Process Definition	supports	Considers mission/business impact in prioritization
PBSAI-A-RSK-007 Risk Acceptance and Exception Handling	RA-7 Risk Response	equivalent	Requires accepted risks and exceptions to be documented and approved
PBSAI-A-RSK-007 Risk Acceptance and Exception Handling	PM-9 Risk Management Strategy	supports	Aligns exceptions with enterprise risk tolerance
PBSAI-A-RSK-007 Risk Acceptance and Exception Handling	AU-12 Audit Record Generation	supports	Records exception approvals and monitoring

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-RSK-008 Risk Feedback and Improvement	PM-14 Testing, Training, and Monitoring	strongly-supports	Improves risk processes based on outcomes and incidents
PBSAI-A-RSK-008 Risk Feedback and Improvement	CA-7 Continuous Monitoring	supports	Uses monitoring outcomes to improve risk models
PBSAI-A-RSK-008 Risk Feedback and Improvement	RA-3 Risk Assessment	enhances	Updates assessment methods based on operational evidence

6. Domain A — Continued

A-CMP — Compliance and Control Validation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-CMP-001 Control-to-Requirement Mapping	CA-2 Control Assessments	equivalent	Maps control activity and evidence to applicable requirements
PBSAI-A-CMP-001 Control-to-Requirement Mapping	PL-2 System Security and Privacy Plans	supports	Links controls to documented system governance obligations
PBSAI-A-CMP-001 Control-to-Requirement Mapping	PM-1 Information Security Program Plan	supports	Supports enterprise control-program organization
PBSAI-A-CMP-002 Compliance Gap Identification	CA-2 Control Assessments	equivalent	Identifies gaps between implemented controls and requirements

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-CMP-002 Compliance Gap Identification	CA-5 Plan of Action and Milestones	strongly-supports	Feeds identified deficiencies into remediation planning
PBSAI-A-CMP-002 Compliance Gap Identification	CA-7 Continuous Monitoring	supports	Detects compliance drift continuously
PBSAI-A-CMP-003 Continuous Compliance Evaluation	CA-7 Continuous Monitoring	equivalent	Continuously evaluates compliance status from system behavior and evidence
PBSAI-A-CMP-003 Continuous Compliance Evaluation	AU-6 Audit Record Review, Analysis, and Reporting	supports	Uses audit records to assess compliance posture
PBSAI-A-CMP-004 Compliance Reporting	CA-2 Control Assessments	supports	Produces structured reports supporting assessment activities
PBSAI-A-CMP-004 Compliance Reporting	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Supports compliance analysis and reporting
PBSAI-A-CMP-004 Compliance Reporting	AU-12 Audit Record Generation	supports	Relies on generated audit records for reporting
PBSAI-A-CMP-005 Evidence-Based Compliance Validation	CA-2 Control Assessments	equivalent	Bases compliance claims on verifiable evidence artifacts
PBSAI-A-CMP-005 Evidence-Based Compliance Validation	AU-3 Content of Audit Records	strongly-supports	Uses structured audit content as validation evidence

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-CMP-005 Evidence-Based Compliance Validation	AU-12 Audit Record Generation	supports	Requires generated records for compliance validation
PBSAI-A-CMP-006 Control Effectiveness Validation	CA-2 Control Assessments	equivalent	Evaluates whether controls operate effectively
PBSAI-A-CMP-006 Control Effectiveness Validation	CA-7 Continuous Monitoring	strongly-supports	Measures ongoing control effectiveness
PBSAI-A-CMP-006 Control Effectiveness Validation	SI-4 System Monitoring	supports	Uses monitoring data to determine effectiveness
PBSAI-A-CMP-007 Regulatory Change Adaptation	PM-14 Testing, Training, and Monitoring	supports	Updates compliance posture based on new requirements and lessons learned
PBSAI-A-CMP-007 Regulatory Change Adaptation	CA-7 Continuous Monitoring	supports	Triggers reassessment when requirements change
PBSAI-A-CMP-007 Regulatory Change Adaptation	PL-2 System Security and Privacy Plans	supports	Updates system plans and mappings as requirements evolve
PBSAI-A-CMP-008 Audit Support and Traceability	AU-6 Audit Record Review, Analysis, and Reporting	equivalent	Supports audit processes through traceable control/evidence relationships
PBSAI-A-CMP-008 Audit Support and Traceability	AU-12 Audit Record Generation	strongly-supports	Maintains records needed to reconstruct compliance assertions

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-CMP-008 Audit Support and Traceability	CA-2 Control Assessments	supports	Supports evidence-based assessment and validation

A-MON — Monitoring, Metrics, and Governance Analytics

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-MON-001 Control Effectiveness Monitoring	CA-7 Continuous Monitoring	equivalent	Continuously monitors the operation and degradation of controls
PBSAI-A-MON-001 Control Effectiveness Monitoring	SI-4 System Monitoring	strongly-supports	Uses monitoring signals to detect control-health problems
PBSAI-A-MON-002 Metrics Definition and Collection	PM-6 Measures of Performance	equivalent	Defines and collects risk, control, and system-health metrics
PBSAI-A-MON-002 Metrics Definition and Collection	CA-7 Continuous Monitoring	supports	Feeds monitoring programs with quantitative metrics
PBSAI-A-MON-003 Governance Dashboards	PM-6 Measures of Performance	strongly-supports	Presents performance and governance metrics to decision-makers
PBSAI-A-MON-003 Governance Dashboards	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports audit and governance review through visibility outputs
PBSAI-A-MON-003 Governance Dashboards	CA-7 Continuous Monitoring	supports	Presents continuous monitoring outputs

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-MON-004 Anomaly Detection in Governance Metrics	SI-4 System Monitoring	strongly-supports	Detects unusual governance/control metric behavior
PBSAI-A-MON-004 Anomaly Detection in Governance Metrics	CA-7 Continuous Monitoring	supports	Enables continuous detection of control degradation
PBSAI-A-MON-005 Feedback into Governance Decisions	PM-14 Testing, Training, and Monitoring	strongly-supports	Uses monitoring findings to improve governance decisions
PBSAI-A-MON-005 Feedback into Governance Decisions	CA-7 Continuous Monitoring	supports	Feeds continuous monitoring outcomes into governance
PBSAI-A-MON-005 Feedback into Governance Decisions	RA-7 Risk Response	supports	Uses monitoring findings to inform response decisions

A-SCRM — Supplier Risk Oversight

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-SCRM-001 Supplier Risk Assessment	SR-3 Supply Chain Controls and Processes	equivalent	Requires supplier and third-party risk assessment
PBSAI-A-SCRM-001 Supplier Risk Assessment	SR-6 Supplier Assessments and Reviews	equivalent	Evaluates supplier security, privacy, and operational posture

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-SCRM-001 Supplier Risk Assessment	RA-3 Risk Assessment	supports	Incorporates suppliers into enterprise risk assessment
PBSAI-A-SCRM-002 Supplier Risk Integration	SR-3 Supply Chain Controls and Processes	strongly-supports	Incorporates supplier risk into governance and procurement decisions
PBSAI-A-SCRM-002 Supplier Risk Integration	SR-5 Acquisition Strategies, Tools, and Methods	supports	Influences acquisition and integration decisions
PBSAI-A-SCRM-002 Supplier Risk Integration	SA-4 Acquisition Process	supports	Uses supplier risk in acquisition decision-making
PBSAI-A-SCRM-003 Supplier Exception Management	SR-3 Supply Chain Controls and Processes	strongly-supports	Documents and approves exceptions to supplier-risk policy
PBSAI-A-SCRM-003 Supplier Exception Management	RA-7 Risk Response	supports	Treats supplier exceptions as accepted or mitigated risk
PBSAI-A-SCRM-003 Supplier Exception Management	AU-12 Audit Record Generation	supports	Records supplier exception approvals and monitoring
PBSAI-A-SCRM-004 Continuous Supplier Monitoring	SR-6 Supplier Assessments and Reviews	strongly-supports	Continuously monitors supplier security and compliance posture
PBSAI-A-SCRM-004 Continuous Supplier Monitoring	CA-7 Continuous Monitoring	supports	Treats supplier posture as part of continuous monitoring
PBSAI-A-SCRM-004 Continuous Supplier Monitoring	SI-5 Security Alerts, Advisories, and Directives	supports	Uses external advisories and threat intelligence for supplier monitoring

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-A-SCRM-005 Supplier Offboarding and Risk Closure	SR-12 Component Disposal	strongly-supports	Ensures supplier relationships and components are securely terminated
PBSAI-A-SCRM-005 Supplier Offboarding and Risk Closure	AC-2 Account Management	supports	Revokes supplier identities and access during offboarding
PBSAI-A-SCRM-005 Supplier Offboarding and Risk Closure	MP-6 Media Sanitization	supports	Supports secure closure, disposal, and data handling

7. Domain B — Asset, Configuration, and Change → NIST SP 800-53 Rev 5

Domain B provides the authoritative operational state layer for PBSAI. It covers asset inventory, configuration baselines, state reconciliation, drift detection, and authoritative state maintenance. The PBSAI Control Catalog describes Domain B as mapping strongly to CM, RA, SI, AU, and CA families.

B-AST — Asset Inventory and State Authority

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-AST-001 Asset Inventory Requirement	CM-8 System Component Inventory	equivalent	Requires authoritative inventory of managed assets
PBSAI-B-AST-001 Asset Inventory Requirement	PM-5 System Inventory	supports	Supports enterprise-level inventory awareness
PBSAI-B-AST-002 Asset Metadata Completeness	CM-8(1) Updates During Installation and Removal	strongly-supports	Maintains metadata needed for inventory correctness and lifecycle tracking

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-AST-002 Asset Metadata Completeness	RA-2 Security Categorization	supports	Provides asset classification and criticality metadata
PBSAI-B-AST-002 Asset Metadata Completeness	AC-16 Security and Privacy Attributes	supports	Maintains ownership, classification, sensitivity, and dependency attributes
PBSAI-B-AST-003 Continuous Asset Discovery	CM-8(2) Automated Maintenance	equivalent	Supports automated and continuous inventory maintenance
PBSAI-B-AST-003 Continuous Asset Discovery	CM-8(3) Automated Unauthorized Component Detection	strongly-supports	Discovers unknown or unauthorized assets
PBSAI-B-AST-003 Continuous Asset Discovery	CA-7 Continuous Monitoring	supports	Feeds continuous monitoring with inventory updates
PBSAI-B-AST-004 Shadow Asset Detection	CM-8(3) Automated Unauthorized Component Detection	equivalent	Detects unmanaged, unauthorized, or shadow assets
PBSAI-B-AST-004 Shadow Asset Detection	RA-5 Vulnerability Monitoring and Scanning	supports	Identifies unmanaged assets that may create vulnerability exposure
PBSAI-B-AST-004 Shadow Asset Detection	SI-4 System Monitoring	supports	Detects unexpected assets through monitoring signals
PBSAI-B-AST-005 Asset Lifecycle Tracking	CM-8 System Component Inventory	strongly-supports	Tracks assets through creation, modification, operation, and retirement
PBSAI-B-AST-005 Asset Lifecycle Tracking	CM-3 Configuration Change Control	supports	Connects asset lifecycle transitions to change governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-AST-005 Asset Lifecycle Tracking	MP-6 Media Sanitization	supports	Supports disposal and end-of-life handling
PBSAI-B-AST-006 Asset Ownership and Accountability	CM-8(1) Updates During Installation and Removal	supports	Maintains ownership metadata as part of inventory
PBSAI-B-AST-006 Asset Ownership and Accountability	PL-2 System Security and Privacy Plans	supports	Links asset ownership to system planning and accountability
PBSAI-B-AST-006 Asset Ownership and Accountability	PM-9 Risk Management Strategy	supports	Assigns accountability for risk ownership
PBSAI-B-AST-007 Asset Dependency Mapping	CM-8(6) Assessed Configurations and Approved Deviations	supports	Documents dependencies and operational relationships
PBSAI-B-AST-007 Asset Dependency Mapping	RA-3 Risk Assessment	strongly-supports	Enables dependency-aware risk analysis
PBSAI-B-AST-007 Asset Dependency Mapping	CP-2 Contingency Plan	supports	Provides dependency data for continuity planning
PBSAI-B-AST-008 Authoritative State Representation	CM-8 System Component Inventory	equivalent	Maintains authoritative operational asset state
PBSAI-B-AST-008 Authoritative State Representation	SI-7 Software, Firmware, and Information Integrity	supports	Supports integrity assessment of system state
PBSAI-B-AST-008 Authoritative State Representation	CA-7 Continuous Monitoring	supports	Provides authoritative baseline for monitoring

B-CFG — Configuration and Baseline Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-CFG-001 Baseline Configuration Definition	CM-2 Baseline Configuration	equivalent	Defines approved baseline configurations
PBSAI-B-CFG-001 Baseline Configuration Definition	CM-6 Configuration Settings	strongly-supports	Establishes secure configuration settings
PBSAI-B-CFG-002 Configuration Drift Detection	CM-6 Configuration Settings	strongly-supports	Detects deviations from approved configuration settings
PBSAI-B-CFG-002 Configuration Drift Detection	SI-7 Software, Firmware, and Information Integrity	supports	Detects unauthorized or unexpected state changes
PBSAI-B-CFG-002 Configuration Drift Detection	CA-7 Continuous Monitoring	supports	Supports continuous configuration monitoring
PBSAI-B-CFG-003 Configuration Change Traceability	CM-3 Configuration Change Control	equivalent	Links configuration changes to requests, identities, and decision context
PBSAI-B-CFG-003 Configuration Change Traceability	AU-12 Audit Record Generation	strongly-supports	Records configuration change evidence
PBSAI-B-CFG-003 Configuration Change Traceability	AC-3 Access Enforcement	supports	Links changes to authorization context
PBSAI-B-CFG-004 Unauthorized Configuration Change Detection	CM-3 Configuration Change Control	strongly-supports	Flags configuration changes lacking authorization

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-CFG-004 Unauthorized Configuration Change Detection	CM-5 Access Restrictions for Change	supports	Detects unauthorized change attempts
PBSAI-B-CFG-004 Unauthorized Configuration Change Detection	SI-4 System Monitoring	supports	Uses monitoring signals to detect unauthorized changes
PBSAI-B-CFG-005 Configuration Enforcement	CM-6 Configuration Settings	equivalent	Enforces approved baseline settings
PBSAI-B-CFG-005 Configuration Enforcement	SI-2 Flaw Remediation	supports	Corrects configuration weaknesses or deviations
PBSAI-B-CFG-005 Configuration Enforcement	CM-3 Configuration Change Control	supports	Ensures enforcement actions are treated as governed changes
PBSAI-B-CFG-006 Configuration Version Control	CM-3 Configuration Change Control	strongly-supports	Maintains versioned history of configuration changes
PBSAI-B-CFG-006 Configuration Version Control	CM-5 Access Restrictions for Change	supports	Supports controlled configuration modification authority
PBSAI-B-CFG-006 Configuration Version Control	SA-10 Developer Configuration Management	supports	Supports version management in development and deployment contexts
PBSAI-B-CFG-007 Pre-Deployment Configuration Validation	SA-10 Developer Configuration Management	strongly-supports	Validates configuration before deployment

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-CFG-007 Pre-Deployment Configuration Validation	SA-11 Developer Testing and Evaluation	supports	Supports testing and evaluation of configuration before release
PBSAI-B-CFG-007 Pre-Deployment Configuration Validation	CM-3 Configuration Change Control	supports	Ensures configuration changes are validated prior to implementation
PBSAI-B-CFG-008 Configuration Scope and Boundary Control	CM-7 Least Functionality	strongly-supports	Limits configuration scope to necessary functionality
PBSAI-B-CFG-008 Configuration Scope and Boundary Control	SC-7 Boundary Protection	supports	Preserves configuration boundaries
PBSAI-B-CFG-008 Configuration Scope and Boundary Control	AC-4 Information Flow Enforcement	supports	Restricts configuration effects across information-flow boundaries

B-REC — Reconciliation and State Integrity

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-REC-001 Multi-Source State Reconciliation	CM-8 System Component Inventory	strongly-supports	Reconciles inventory and state information across sources
PBSAI-B-REC-001 Multi-Source State Reconciliation	CA-7 Continuous Monitoring	supports	Uses monitoring and inventory signals to maintain current state
PBSAI-B-REC-002 Conflict Resolution	SI-10 Information Input Validation	strongly-supports	Resolves conflicting or invalid state information deterministically

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-REC-002 Conflict Resolution	CM-8 System Component Inventory	supports	Ensures inventory consistency
PBSAI-B-REC-002 Conflict Resolution	AU-12 Audit Record Generation	supports	Records state conflicts and resolutions
PBSAI-B-REC-003 Authoritative State Maintenance	CM-8 System Component Inventory	equivalent	Maintains authoritative asset and configuration state
PBSAI-B-REC-003 Authoritative State Maintenance	SI-7 Software, Firmware, and Information Integrity	strongly-supports	Supports integrity of operational state
PBSAI-B-REC-004 State Integrity Validation	SI-7 Software, Firmware, and Information Integrity	equivalent	Validates correctness and integrity of reconciled system state
PBSAI-B-REC-004 State Integrity Validation	CA-7 Continuous Monitoring	supports	Continuously evaluates state integrity
PBSAI-B-REC-005 Reconciliation Auditability	AU-3 Content of Audit Records	strongly-supports	Records reconciliation inputs, conflicts, and decisions
PBSAI-B-REC-005 Reconciliation Auditability	AU-12 Audit Record Generation	equivalent	Generates audit records for reconciliation operations
PBSAI-B-REC-006 State Drift Detection	CM-6 Configuration Settings	strongly-supports	Detects drift from approved configuration or state
PBSAI-B-REC-006 State Drift Detection	SI-7 Software, Firmware, and Information Integrity	strongly-supports	Detects unauthorized integrity changes
PBSAI-B-REC-006 State Drift Detection	SI-4 System Monitoring	supports	Uses monitoring signals to detect state drift

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-B-REC-007 Reconciliation Feedback Loop	CA-7 Continuous Monitoring	supports	Feeds reconciliation findings into continuous monitoring
PBSAI-B-REC-007 Reconciliation Feedback Loop	RA-7 Risk Response	supports	Uses state reconciliation findings to trigger response decisions
PBSAI-B-REC-007 Reconciliation Feedback Loop	SI-4 System Monitoring	supports	Improves monitoring based on reconciliation outcomes

8. Domain C — Identity, Credential, and Access Management → NIST SP 800-53 Rev 5

Domain C covers identity lifecycle management, authorization, credential management, identity graphing, and identity evidence/attribution. PBSAI's Domain C mapping is strongest against AC, IA, and AU control families.

C-IAM — Identity Lifecycle Management

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-IAM-001 Identity Provisioning	AC-2 Account Management	equivalent	Establishes creation and management of identities
PBSAI-C-IAM-001 Identity Provisioning	IA-2 Identification and Authentication	supports	Provides identities used for authentication
PBSAI-C-IAM-001 Identity Provisioning	IA-4 Identifier Management	strongly- supports	Assigns and governs identity identifiers

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-IAM-002 Identity Attribute Management	AC-2 Account Management	strongly-supports	Maintains identity account attributes
PBSAI-C-IAM-002 Identity Attribute Management	AC-16 Security and Privacy Attributes	equivalent	Maintains identity attributes used in access and governance decisions
PBSAI-C-IAM-002 Identity Attribute Management	IA-4 Identifier Management	supports	Maintains identifier-associated metadata
PBSAI-C-IAM-003 Identity Deprovisioning	AC-2 Account Management	equivalent	Removes accounts and identities when no longer authorized
PBSAI-C-IAM-003 Identity Deprovisioning	IA-4 Identifier Management	supports	Retires identifiers associated with deprovisioned identities
PBSAI-C-IAM-003 Identity Deprovisioning	PS-4 Personnel Termination	supports	Supports termination-related access removal
PBSAI-C-IAM-004 Identity Activity Logging	AU-2 Event Logging	equivalent	Logs identity-related events
PBSAI-C-IAM-004 Identity Activity Logging	AU-3 Content of Audit Records	strongly-supports	Preserves identity event details
PBSAI-C-IAM-004 Identity Activity Logging	AU-12 Audit Record Generation	equivalent	Generates audit records for identity lifecycle events
PBSAI-C-IAM-005 Identity Validation	AC-3 Access Enforcement	strongly-supports	Validates identity prior to access or action authorization

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-IAM-005 Identity Validation	IA-2 Identification and Authentication	equivalent	Validates claimed identity
PBSAI-C-IAM-005 Identity Validation	IA-5 Authenticator Management	supports	Uses managed credentials/authenticators during validation
PBSAI-C-IAM-006 Identity Uniqueness	IA-4 Identifier Management	equivalent	Ensures unique identification of entities
PBSAI-C-IAM-006 Identity Uniqueness	IA-2 Identification and Authentication	supports	Supports reliable authentication through unique identities
PBSAI-C-IAM-007 Federated Identity Integration	IA-8 Identification and Authentication — Non-Organizational Users	strongly-supports	Supports external identity provider integration
PBSAI-C-IAM-007 Federated Identity Integration	AC-20 Use of External Systems	supports	Governs identity use across external systems
PBSAI-C-IAM-007 Federated Identity Integration	SR-3 Supply Chain Controls and Processes	supports	Addresses third-party identity dependencies
PBSAI-C-IAM-008 Identity Lifecycle Monitoring	AC-2 Account Management	strongly-supports	Monitors account state changes
PBSAI-C-IAM-008 Identity Lifecycle Monitoring	IA-4 Identifier Management	supports	Tracks identity identifier lifecycle
PBSAI-C-IAM-008 Identity Lifecycle Monitoring	CA-7 Continuous Monitoring	supports	Continuously monitors identity lifecycle posture

C-AUTH — Authorization and Access Control

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-AUTH-001 Least Privilege Enforcement	AC-6 Least Privilege	equivalent	Enforces minimum necessary access rights
PBSAI-C-AUTH-001 Least Privilege Enforcement	AC-3 Access Enforcement	strongly-supports	Applies least-privilege constraints to access decisions
PBSAI-C-AUTH-002 Separation of Duties	AC-5 Separation of Duties	equivalent	Prevents conflicting privileges and authority combinations
PBSAI-C-AUTH-002 Separation of Duties	AC-6 Least Privilege	supports	Restricts excessive privilege accumulation
PBSAI-C-AUTH-003 Contextual Authorization	AC-3 Access Enforcement	equivalent	Evaluates access based on identity, policy, context, and risk
PBSAI-C-AUTH-003 Contextual Authorization	AC-6 Least Privilege	supports	Applies context-aware least privilege
PBSAI-C-AUTH-003 Contextual Authorization	RA-3 Risk Assessment	supports	Incorporates risk context into authorization decisions
PBSAI-C-AUTH-004 Attribute-Based Access Control	AC-16 Security and Privacy Attributes	equivalent	Uses attributes to drive authorization decisions
PBSAI-C-AUTH-004 Attribute-Based Access Control	AC-3 Access Enforcement	strongly-supports	Enforces decisions based on security/privacy attributes

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-AUTH-004 Attribute-Based Access Control	IA-2 Identification and Authentication	supports	Relies on authenticated identities and attributes
PBSAI-C-AUTH-005 Privileged Access Management	AC-6(2) Non-Privileged Access for Nonsecurity Functions	supports	Restricts privileged functions and access
PBSAI-C-AUTH-005 Privileged Access Management	AC-2 Account Management	strongly-supports	Governs privileged account assignment and lifecycle
PBSAI-C-AUTH-005 Privileged Access Management	IA-11 Re-Authentication	supports	Supports stronger validation for privileged activity
PBSAI-C-AUTH-006 Just-in-Time Access	AC-2 Account Management	strongly-supports	Enables temporary access provisioning and removal
PBSAI-C-AUTH-006 Just-in-Time Access	AC-6 Least Privilege	enhances	Reduces standing privilege through time-bounded access
PBSAI-C-AUTH-007 Access Enforcement Monitoring	AC-3 Access Enforcement	supports	Monitors whether access enforcement operates correctly
PBSAI-C-AUTH-007 Access Enforcement Monitoring	AC-7 Unsuccessful Logon Attempts	supports	Detects abnormal access behavior
PBSAI-C-AUTH-007 Access Enforcement Monitoring	SI-4 System Monitoring	strongly-supports	Monitors access behavior and enforcement outcomes
PBSAI-C-AUTH-008 Access Decision Logging	AU-3 Content of Audit Records	equivalent	Records authorization decision details

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-AUTH-008 Access Decision Logging	AU-12 Audit Record Generation	equivalent	Generates audit records for authorization decisions
PBSAI-C-AUTH-008 Access Decision Logging	AC-3 Access Enforcement	supports	Provides evidence of access enforcement
PBSAI-C-AUTH-009 Policy-Based Access Enforcement	AC-3 Access Enforcement	equivalent	Enforces access control based on policy
PBSAI-C-AUTH-009 Policy-Based Access Enforcement	AC-4 Information Flow Enforcement	supports	Applies policy to information flows where relevant
PBSAI-C-AUTH-009 Policy-Based Access Enforcement	PL-2 System Security and Privacy Plans	supports	Links access enforcement to documented policy
PBSAI-C-AUTH-010 Risk-Adaptive Access Control	AC-3 Access Enforcement	enhances	Adjusts access decisions using runtime risk
PBSAI-C-AUTH-010 Risk-Adaptive Access Control	AC-16 Security and Privacy Attributes	supports	Uses contextual security attributes in adaptive access
PBSAI-C-AUTH-010 Risk-Adaptive Access Control	RA-3 Risk Assessment	strongly-supports	Incorporates risk assessment into access decisions

C-CRED — Credential Management

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-CRED-001 Credential Issuance	IA-5 Authenticator Management	equivalent	Governs issuance and management of authenticators
PBSAI-C-CRED-001 Credential Issuance	IA-2 Identification and Authentication	supports	Provides credentials used for authentication
PBSAI-C-CRED-002 Credential Rotation	IA-5 Authenticator Management	equivalent	Requires credential update and lifecycle management
PBSAI-C-CRED-002 Credential Rotation	IA-5(1) Password-Based Authentication	supports	Supports periodic rotation where password-based credentials are used
PBSAI-C-CRED-003 Credential Revocation	IA-5 Authenticator Management	strongly-supports	Removes invalid or compromised credentials
PBSAI-C-CRED-003 Credential Revocation	AC-2 Account Management	supports	Supports account disablement and access removal
PBSAI-C-CRED-003 Credential Revocation	IR-4 Incident Handling	supports	Enables incident-driven credential revocation
PBSAI-C-CRED-004 Credential Protection	IA-5 Authenticator Management	equivalent	Protects authenticators from unauthorized disclosure or misuse
PBSAI-C-CRED-004 Credential Protection	SC-12 Cryptographic Key Establishment and Management	supports	Applies cryptographic protections for credential/key material
PBSAI-C-CRED-004 Credential Protection	SC-13 Cryptographic Protection	supports	Protects credentials using cryptographic mechanisms

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-CRED-005 Multi-Factor Authentication	IA-2(1) Multi-Factor Authentication to Privileged Accounts	equivalent	Requires MFA for privileged access
PBSAI-C-CRED-005 Multi-Factor Authentication	IA-2(2) Multi-Factor Authentication to Non-Privileged Accounts	strongly-supports	Supports MFA for ordinary users
PBSAI-C-CRED-005 Multi-Factor Authentication	IA-2(12) Acceptance of PIV Credentials	supports	Supports strong credential use where applicable
PBSAI-C-CRED-006 Credential Scope Limitation	IA-5 Authenticator Management	strongly-supports	Restricts credential use to approved contexts
PBSAI-C-CRED-006 Credential Scope Limitation	AC-6 Least Privilege	supports	Limits credential authority to least necessary scope
PBSAI-C-CRED-007 Credential Monitoring	SI-4 System Monitoring	strongly-supports	Monitors credential use and anomalies
PBSAI-C-CRED-007 Credential Monitoring	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports review of credential-related events
PBSAI-C-CRED-007 Credential Monitoring	IA-5 Authenticator Management	supports	Detects credential misuse and compromise

C-GRAPH — Identity Relationships and Context Graph

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-GRAPH-001 Identity Relationship Modeling	AC-2 Account Management	supports	Models relationships among identities, roles, and resources
PBSAI-C-GRAPH-001 Identity Relationship Modeling	AC-16 Security and Privacy Attributes	strongly-supports	Provides relationship attributes for governance decisions
PBSAI-C-GRAPH-001 Identity Relationship Modeling	PM-12 Insider Threat Program	supports	Supports insider-risk correlation and relationship analysis
PBSAI-C-GRAPH-002 Identity Traceability	AU-10 Non-Repudiation	equivalent	Enables attribution of actions to identities
PBSAI-C-GRAPH-002 Identity Traceability	AU-3 Content of Audit Records	strongly-supports	Records identity/action relationship data
PBSAI-C-GRAPH-002 Identity Traceability	IA-2 Identification and Authentication	supports	Links traceability to authenticated identity
PBSAI-C-GRAPH-003 Cross-Domain Correlation	AC-16 Security and Privacy Attributes	supports	Correlates identity attributes across domains
PBSAI-C-GRAPH-003 Cross-Domain Correlation	AU-12 Audit Record Generation	supports	Supports correlated audit records
PBSAI-C-GRAPH-003 Cross-Domain Correlation	SI-4 System Monitoring	strongly-supports	Enables identity-aware monitoring across systems
PBSAI-C-GRAPH-004 Delegation Tracking	AC-3 Access Enforcement	supports	Records delegated authority relevant to access decisions
PBSAI-C-GRAPH-004 Delegation Tracking	AC-6 Least Privilege	supports	Prevents excessive delegated privilege

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-GRAPH-004 Delegation Tracking	AU-10 Non-Repudiation	strongly-supports	Maintains attribution through delegated authority chains

C-EVID — Identity Evidence and Attribution

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-EVID-001 Identity Evidence Collection	AU-2 Event Logging	equivalent	Collects identity-related events
PBSAI-C-EVID-001 Identity Evidence Collection	AU-3 Content of Audit Records	strongly-supports	Preserves identity-specific evidence content
PBSAI-C-EVID-001 Identity Evidence Collection	AU-12 Audit Record Generation	equivalent	Generates identity evidence records
PBSAI-C-EVID-002 Identity Evidence Correlation	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Correlates identity evidence across events
PBSAI-C-EVID-002 Identity Evidence Correlation	AU-12 Audit Record Generation	supports	Links related identity records
PBSAI-C-EVID-003 Non-Repudiation Enforcement	AU-10 Non-Repudiation	equivalent	Ensures actions can be attributed to identities
PBSAI-C-EVID-003 Non-Repudiation Enforcement	IA-5 Authenticator Management	supports	Uses credential integrity to support attribution

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-C-EVID-003 Non-Repudiation Enforcement	SI-7 Software, Firmware, and Information Integrity	supports	Protects attribution-related records and mechanisms
PBSAI-C-EVID-004 Identity Evidence Integrity	AU-9 Protection of Audit Information	equivalent	Protects identity evidence from unauthorized modification
PBSAI-C-EVID-004 Identity Evidence Integrity	SI-7 Software, Firmware, and Information Integrity	strongly-supports	Preserves integrity of identity evidence artifacts

9. Domain D — Threat Intelligence, Monitoring, and Detection → NIST SP 800-53 Rev 5

Domain D covers telemetry, threat intelligence, detection, UEBA, threat hunting, exposure mapping, and deception. PBSAI's Domain D mapping is strongest against SI, RA, IR, AU, and CA families.

D-MON — Telemetry and Observability

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-MON-001 Telemetry Collection	AU-2 Event Logging	equivalent	Captures system and security-relevant events
PBSAI-D-MON-001 Telemetry Collection	AU-12 Audit Record Generation	strongly-supports	Produces telemetry records usable as audit evidence
PBSAI-D-MON-001 Telemetry Collection	SI-4 System Monitoring	equivalent	Collects monitoring data from systems and environments

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-MON-002 Telemetry Normalization	AU-3 Content of Audit Records	strongly-supports	Structures telemetry for analysis and auditability
PBSAI-D-MON-002 Telemetry Normalization	AU-12 Audit Record Generation	supports	Ensures generated telemetry is analyzable and consistent
PBSAI-D-MON-003 Event Correlation	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Correlates security events across systems
PBSAI-D-MON-003 Event Correlation	SI-4 System Monitoring	equivalent	Supports cross-source monitoring and detection
PBSAI-D-MON-004 Telemetry Integrity Protection	AU-9 Protection of Audit Information	equivalent	Protects telemetry and audit data from tampering
PBSAI-D-MON-004 Telemetry Integrity Protection	SI-7 Software, Firmware, and Information Integrity	strongly-supports	Preserves integrity of telemetry artifacts
PBSAI-D-MON-005 Telemetry Retention	AU-11 Audit Record Retention	equivalent	Retains telemetry for audit, investigation, and analysis
PBSAI-D-MON-005 Telemetry Retention	AU-9 Protection of Audit Information	supports	Protects retained telemetry
PBSAI-D-MON-006 Observability Coverage	CA-7 Continuous Monitoring	strongly-supports	Ensures sufficient visibility for ongoing assessment
PBSAI-D-MON-006 Observability Coverage	SI-4 System Monitoring	equivalent	Ensures systems are monitored across relevant coverage areas

D-INT — Threat Intelligence

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-INT-001 Threat Intelligence Ingestion	SI-5 Security Alerts, Advisories, and Directives	equivalent	Ingests threat feeds, advisories, and intelligence
PBSAI-D-INT-001 Threat Intelligence Ingestion	RA-5 Vulnerability Monitoring and Scanning	supports	Supplies intelligence relevant to vulnerabilities and exposures
PBSAI-D-INT-002 Threat Intelligence Validation	SI-7 Software, Firmware, and Information Integrity	supports	Validates integrity and trustworthiness of intelligence artifacts
PBSAI-D-INT-002 Threat Intelligence Validation	SI-5 Security Alerts, Advisories, and Directives	strongly-supports	Ensures consumed intelligence is relevant and reliable
PBSAI-D-INT-003 Threat Intelligence Normalization	SI-5 Security Alerts, Advisories, and Directives	supports	Structures intelligence for downstream use
PBSAI-D-INT-003 Threat Intelligence Normalization	AU-3 Content of Audit Records	supports	Provides structured intelligence records
PBSAI-D-INT-004 Threat Intelligence Integration	SI-4 System Monitoring	strongly-supports	Feeds intelligence into monitoring and detection
PBSAI-D-INT-004 Threat Intelligence Integration	CA-7 Continuous Monitoring	supports	Supports continuous monitoring using updated threat context
PBSAI-D-INT-005 Threat Intelligence Prioritization	RA-5 Vulnerability Monitoring and Scanning	strongly-supports	Prioritizes vulnerabilities and indicators using intelligence
PBSAI-D-INT-005 Threat Intelligence Prioritization	RA-3 Risk Assessment	supports	Feeds intelligence into risk scoring

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-INT-006 Threat Intelligence Lifecycle Management	CA-7 Continuous Monitoring	supports	Maintains currency and relevance of intelligence
PBSAI-D-INT-006 Threat Intelligence Lifecycle Management	PM-16 Threat Awareness Program	strongly-supports	Supports threat awareness through managed intelligence lifecycle

D-DET — Detection and Analytics

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-DET-001 Detection Rule Definition	SI-4 System Monitoring	equivalent	Defines rules for identifying threats and anomalies
PBSAI-D-DET-001 Detection Rule Definition	SI-3 Malicious Code Protection	supports	Supports detection of malicious code behavior
PBSAI-D-DET-002 Detection Rule Execution	SI-4 System Monitoring	equivalent	Executes detection logic against telemetry
PBSAI-D-DET-002 Detection Rule Execution	IR-5 Incident Monitoring	supports	Feeds incident monitoring with detection outputs
PBSAI-D-DET-003 Alert Generation	SI-4 System Monitoring	strongly-supports	Produces alerts from detection events
PBSAI-D-DET-003 Alert Generation	IR-5 Incident Monitoring	supports	Supports incident monitoring and triage
PBSAI-D-DET-003 Alert Generation	AU-6 Audit Record Review, Analysis, and Reporting	supports	Enables alert review and analysis

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-DET-004 Detection Tuning and Optimization	CA-7 Continuous Monitoring	supports	Improves detection over time through monitoring feedback
PBSAI-D-DET-004 Detection Tuning and Optimization	SI-4 System Monitoring	enhances	Tunes and improves detection effectiveness
PBSAI-D-DET-005 False Positive/Negative Management	CA-7 Continuous Monitoring	supports	Improves monitoring effectiveness
PBSAI-D-DET-005 False Positive/Negative Management	AU-6 Audit Record Review, Analysis, and Reporting	supports	Uses review outcomes to reduce detection error

D-UEBA — User and Entity Behavior Analytics

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-UEBA-001 Behavioral Baseline Modeling	SI-4 System Monitoring	enhances	Establishes expected behavior patterns for anomaly detection
PBSAI-D-UEBA-001 Behavioral Baseline Modeling	AC-2 Account Management	supports	Uses account activity to build identity behavior models
PBSAI-D-UEBA-001 Behavioral Baseline Modeling	PM-12 Insider Threat Program	supports	Supports insider-risk detection
PBSAI-D-UEBA-002 Behavioral Anomaly Detection	SI-4 System Monitoring	equivalent	Detects deviations from expected behavior

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-UEBA-002 Behavioral Anomaly Detection	IR-5 Incident Monitoring	supports	Feeds anomaly events into incident monitoring
PBSAI-D-UEBA-003 Behavioral Risk Scoring	RA-3 Risk Assessment	strongly-supports	Assigns behavioral risk scores
PBSAI-D-UEBA-003 Behavioral Risk Scoring	SI-4 System Monitoring	supports	Uses monitoring data to produce risk scores

D-HUNT — Threat Hunting

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-D-HUNT-001 Proactive Threat Hunting	SI-4 System Monitoring	enhances	Proactively searches for hidden or emerging threats
PBSAI-D-HUNT-001 Proactive Threat Hunting	RA-5 Vulnerability Monitoring and Scanning	supports	Identifies exploitable patterns and exposures
PBSAI-D-HUNT-001 Proactive Threat Hunting	IR-5 Incident Monitoring	supports	Supports detection and investigation of incidents
PBSAI-D-HUNT-002 Hunt Feedback Integration	CA-7 Continuous Monitoring	supports	Feeds hunt findings into continuous monitoring
PBSAI-D-HUNT-002 Hunt Feedback Integration	RA-7 Risk Response	supports	Uses hunt results to guide response and mitigation

PBSAI Control	NIST SP 800-53 Control	Relationship Rationale	
PBSAI-D-HUNT-002 Hunt Feedback Integration	SI-4 System Monitoring	enhances	Improves monitoring logic from hunt findings

Domain E — Protection, Enforcement, and Response → NIST SP 800-53 Rev 5

Domain E defines PBSAI’s active protection, enforcement, response orchestration, containment, and remediation capabilities. The domain operationalizes policy-bound preventive controls, response execution, corrective action management, bounded automation, and governed state restoration.

Domain E aligns strongly with:

- AC (Access Control),
- SC (System and Communications Protection),
- SI (System and Information Integrity),
- IR (Incident Response),
- CM (Configuration Management),
- RA (Risk Assessment),
- CA (Assessment and Monitoring),
- and CP (Contingency Planning).

The domain is execution-oriented and represents one of PBSAI’s primary operational enforcement layers.

E-PRO — Preventive Protection Controls

PBSAI Control	NIST SP 800-53 Control	Relationship Rationale	
PBSAI-E-PRO-001 Preventive Control Definition	SC-7 Boundary Protection	equivalent	Establishes preventive enforcement boundaries for systems, workloads, and network segments

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-PRO-001 Preventive Control Definition	AC-3 Access Enforcement	strongly- supports	Enforces governed operational restrictions prior to execution
PBSAI-E-PRO-001 Preventive Control Definition	SI-4 System Monitoring	supports	Enables monitoring-informed preventive controls
PBSAI-E-PRO-002 Segmentation and Isolation Enforcement	SC-7 Boundary Protection	equivalent	Implements segmentation and controlled communication boundaries
PBSAI-E-PRO-002 Segmentation and Isolation Enforcement	SC-32 System Partitioning	strongly- supports	Enforces logical isolation between protected operational domains
PBSAI-E-PRO-002 Segmentation and Isolation Enforcement	AC-4 Information Flow Enforcement	supports	Restricts unauthorized flows between domains
PBSAI-E-PRO-003 Endpoint and Workload Protection	SI-3 Malicious Code Protection	equivalent	Applies preventive protections to endpoints and workloads
PBSAI-E-PRO-003 Endpoint and Workload Protection	SI-4 System Monitoring	strongly- supports	Detects unsafe or malicious operational conditions
PBSAI-E-PRO-003 Endpoint and Workload Protection	CM-7 Least Functionality	supports	Restricts unnecessary operational capability
PBSAI-E-PRO-004 Protected Data Access Enforcement	AC-3 Access Enforcement	equivalent	Restricts access to protected data based on policy and context
PBSAI-E-PRO-004 Protected Data Access Enforcement	AC-6 Least Privilege	strongly- supports	Limits access scope and authority

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-PRO-004 Protected Data Access Enforcement	SC-28 Protection of Information at Rest	supports	Helps prevent unauthorized data exposure
PBSAI-E-PRO-005 Cloud and Platform Guardrails	CM-6 Configuration Settings	strongly-supports	Enforces approved cloud and platform security settings
PBSAI-E-PRO-005 Cloud and Platform Guardrails	SC-7 Boundary Protection	supports	Maintains controlled cloud/network boundaries
PBSAI-E-PRO-005 Cloud and Platform Guardrails	SI-7 Software, Firmware, and Information Integrity	supports	Protects integrity of managed platform states
PBSAI-E-PRO-006 Preventive Control Health Monitoring	CA-7 Continuous Monitoring	equivalent	Continuously evaluates preventive-control operational state
PBSAI-E-PRO-006 Preventive Control Health Monitoring	SI-4 System Monitoring	strongly-supports	Monitors control degradation and operational anomalies
PBSAI-E-PRO-006 Preventive Control Health Monitoring	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports operational review and governance reporting

E-ENF — Enforcement and Response Orchestration

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-ENF-001 Response Action Execution	IR-4 Incident Handling	equivalent	Executes governed response and containment actions

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-ENF-001 Response Action Execution	AC-3 Access Enforcement	supports	Prevents unauthorized operational actions during response
PBSAI-E-ENF-001 Response Action Execution	SC-7 Boundary Protection	supports	Applies containment boundaries during response execution
PBSAI-E-ENF-002 Response Authorization Validation	AC-3 Access Enforcement	equivalent	Requires authorization before execution of response actions
PBSAI-E-ENF-002 Response Authorization Validation	AC-5 Separation of Duties	strongly-supports	Separates detection, recommendation, and authorization functions
PBSAI-E-ENF-002 Response Authorization Validation	AU-10 Non-Repudiation	supports	Records attributable authorization decisions
PBSAI-E-ENF-003 Response Scope Constraint	AC-6 Least Privilege	equivalent	Restricts response authority to approved operational scope
PBSAI-E-ENF-003 Response Scope Constraint	SC-7 Boundary Protection	strongly-supports	Prevents response actions from exceeding authorized boundaries
PBSAI-E-ENF-003 Response Scope Constraint	CM-3 Configuration Change Control	supports	Governs controlled operational changes during response
PBSAI-E-ENF-004 Response Execution Logging	AU-2 Event Logging	equivalent	Requires logging of all response activities

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-ENF-004 Response Execution Logging	AU-3 Content of Audit Records	strongly-supports	Captures execution details, actors, timestamps, and outcomes
PBSAI-E-ENF-004 Response Execution Logging	AU-12 Audit Record Generation	equivalent	Produces structured response evidence artifacts
PBSAI-E-ENF-005 Response Outcome Verification	IR-4 Incident Handling	strongly-supports	Verifies whether containment and corrective actions succeeded
PBSAI-E-ENF-005 Response Outcome Verification	CA-7 Continuous Monitoring	supports	Monitors post-response operational state
PBSAI-E-ENF-005 Response Outcome Verification	SI-4 System Monitoring	supports	Detects residual compromise or operational anomalies

E-RMD — Remediation and Recovery Controls

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-RMD-001 Remediation Planning	SI-2 Flaw Remediation	equivalent	Requires planning and coordination of corrective actions
PBSAI-E-RMD-001 Remediation Planning	RA-7 Risk Response	supports	Aligns remediation with operational risk priorities
PBSAI-E-RMD-001 Remediation Planning	CM-3 Configuration Change Control	supports	Treats remediation as governed operational change

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-RMD-002 Remediation Prioritization	RA-7 Risk Response	equivalent	Prioritizes corrective actions based on risk and impact
PBSAI-E-RMD-002 Remediation Prioritization	PM-9 Risk Management Strategy	strongly-supports	Aligns remediation prioritization with enterprise risk tolerance
PBSAI-E-RMD-002 Remediation Prioritization	PM-11 Mission and Business Process Definition	supports	Considers mission and operational criticality
PBSAI-E-RMD-003 Corrective Action Execution	SI-2 Flaw Remediation	equivalent	Executes remediation and corrective operational changes
PBSAI-E-RMD-003 Corrective Action Execution	CM-3 Configuration Change Control	strongly-supports	Governs operational modifications during remediation
PBSAI-E-RMD-003 Corrective Action Execution	CP-10 System Recovery and Reconstitution	supports	Restores systems to approved operational state
PBSAI-E-RMD-004 Remediation Tracking and Status	CA-7 Continuous Monitoring	strongly-supports	Tracks remediation progress and unresolved issues
PBSAI-E-RMD-004 Remediation Tracking and Status	AU-12 Audit Record Generation	supports	Produces remediation evidence and lifecycle tracking
PBSAI-E-RMD-004 Remediation Tracking and Status	PM-6 Measures of Performance	supports	Supports governance metrics and remediation analytics
PBSAI-E-RMD-005 Remediation Validation	SI-2 Flaw Remediation	strongly-supports	Verifies remediation effectiveness before closure

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-E-RMD-005 Remediation Validation	CA-2 Control Assessments	supports	Confirms restored control effectiveness
PBSAI-E-RMD-005 Remediation Validation	SI-4 System Monitoring	supports	Monitors for recurrence or residual operational issues

Domain E — Cross-Domain Governance Alignment

Domain E functions as PBSAI's primary operational enforcement and corrective-action layer.

The domain:

- converts policy and risk decisions into bounded operational action,
- constrains autonomous response execution,
- enforces least-privilege operational scope,
- and preserves auditability and explainability throughout response workflows.

Domain E relies heavily upon:

- Domain A governance and risk outputs,
- Domain C authorization decisions,
- Domain D detection and intelligence signals,
- Domain G incident-handling workflows,
- and Domain H recovery and continuity functions.

The domain also operationalizes several global PBSAI governance controls:

- PBSAI-GEN-MCP (context binding),
- PBSAI-GEN-EVID (evidence generation),
- PBSAI-GEN-HITL (human authorization),
- PBSAI-GEN-SAFE (safe execution and rollback),

- and PBSAI-GEN-PROV (provenance and attribution).

Together, these controls provide:

- explainable response execution,
- bounded autonomous remediation,
- deterministic operational governance,
- and evidence-centric security operations suitable for high-assurance enterprise AI environments.

Domain F — Data Security and Privacy → NIST SP 800-53 Rev 5

Domain F defines PBSAI’s data discovery, classification, protection, privacy, integrity, cryptographic enforcement, retention, disposal, and data-evidence controls.

Domain F aligns strongly with:

- AC — Access Control
- AU — Audit and Accountability
- MP — Media Protection
- PT — Personally Identifiable Information Processing and Transparency
- RA — Risk Assessment
- SC — System and Communications Protection
- SI — System and Information Integrity

F-CLS — Data Discovery and Classification

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-CLS-001 Data Discovery	RA-2 Security Categorization	strongly-supports	Identifies data assets requiring categorization and protection
PBSAI-F-CLS-001 Data Discovery	CM-8 System Component Inventory	supports	Extends inventory visibility to data-bearing assets and repositories

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-CLS-002 Data Classification	AC-16 Security and Privacy Attributes	equivalent	Assigns classification and handling attributes to data assets
PBSAI-F-CLS-002 Data Classification	RA-2 Security Categorization	strongly-supports	Supports impact-aware security categorization
PBSAI-F-CLS-003 Classification Validation	SI-10 Information Input Validation	supports	Validates classification accuracy and consistency
PBSAI-F-CLS-003 Classification Validation	CA-7 Continuous Monitoring	supports	Enables ongoing validation of data classification state
PBSAI-F-CLS-004 Classification Metadata Propagation	AC-16 Security and Privacy Attributes	equivalent	Propagates classification attributes into downstream governance decisions
PBSAI-F-CLS-004 Classification Metadata Propagation	AU-12 Audit Record Generation	supports	Preserves classification context in audit records
PBSAI-F-CLS-005 Classification-Based Policy Binding	AC-3 Access Enforcement	strongly-supports	Applies access and handling policies based on classification
PBSAI-F-CLS-005 Classification-Based Policy Binding	AC-6 Least Privilege	supports	Restricts access based on data sensitivity and need-to-know

F-DLP — Data Loss Prevention and Movement Control

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-DLP-001 Data Movement Monitoring	SI-4 System Monitoring	equivalent	Monitors data movement across systems, users, and channels
PBSAI-F-DLP-001 Data Movement Monitoring	AU-6 Audit Record Review, Analysis, and Reporting	supports	Enables review of suspicious or policy-relevant data movement
PBSAI-F-DLP-002 Unauthorized Data Movement Detection	AC-4 Information Flow Enforcement	strongly-supports	Detects flows that violate permitted data movement policy
PBSAI-F-DLP-002 Unauthorized Data Movement Detection	SI-4 System Monitoring	strongly-supports	Identifies anomalous or unauthorized transfer behavior
PBSAI-F-DLP-003 Data Movement Blocking and Containment	AC-4 Information Flow Enforcement	equivalent	Blocks or constrains prohibited data flows
PBSAI-F-DLP-003 Data Movement Blocking and Containment	IR-4 Incident Handling	supports	Triggers containment workflows for suspected exfiltration
PBSAI-F-DLP-004 High-Impact DLP Escalation	AC-3 Access Enforcement	supports	Requires authorization before high-impact data-control action
PBSAI-F-DLP-004 High-Impact DLP Escalation	AU-12 Audit Record Generation	supports	Records escalation, approval, and enforcement outcomes
PBSAI-F-DLP-005 DLP Evidence Generation	AU-3 Content of Audit Records	strongly-supports	Captures data movement context, actor, channel, policy, and outcome
PBSAI-F-DLP-005 DLP Evidence Generation	IR-5 Incident Monitoring	supports	Supports investigation of potential data-loss incidents

F-CRY — Cryptographic Protection

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-CRY-001 Encryption at Rest and in Transit	SC-13 Cryptographic Protection	equivalent	Enforces approved cryptographic protection for sensitive data
PBSAI-F-CRY-001 Encryption at Rest and in Transit	SC-8 Transmission Confidentiality and Integrity	strongly-supports	Protects data while transmitted across networks
PBSAI-F-CRY-001 Encryption at Rest and in Transit	SC-28 Protection of Information at Rest	strongly-supports	Protects stored sensitive data
PBSAI-F-CRY-002 Cryptographic Configuration Compliance	SC-12 Cryptographic Key Establishment and Management	supports	Ensures cryptographic configuration aligns with policy and key-management requirements
PBSAI-F-CRY-002 Cryptographic Configuration Compliance	CM-6 Configuration Settings	strongly-supports	Validates crypto settings against approved baselines
PBSAI-F-CRY-003 Weak Cryptography Detection	RA-5 Vulnerability Monitoring and Scanning	supports	Identifies weak or deprecated cryptographic mechanisms
PBSAI-F-CRY-003 Weak Cryptography Detection	SI-7 Software, Firmware, and Information Integrity	supports	Supports integrity assurance by preventing weak crypto usage
PBSAI-F-CRY-004 Cryptographic Remediation	RA-7 Risk Response	supports	Drives remediation of weak or noncompliant cryptographic controls
PBSAI-F-CRY-004 Cryptographic Remediation	CM-3 Configuration Change Control	supports	Treats crypto changes as governed configuration changes

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-CRY-005 Crypto Evidence and Attestation	AU-12 Audit Record Generation	supports	Records cryptographic posture, violations, and remediation evidence
PBSAI-F-CRY-005 Crypto Evidence and Attestation	AU-9 Protection of Audit Information	supports	Protects records supporting cryptographic compliance claims

F-INT — Data Integrity Monitoring

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-INT-001 Data Integrity Monitoring	SI-7 Software, Firmware, and Information Integrity	equivalent	Monitors data and information artifacts for unauthorized modification
PBSAI-F-INT-001 Data Integrity Monitoring	AU-9 Protection of Audit Information	supports	Protects integrity-relevant records and evidence
PBSAI-F-INT-002 Integrity Violation Detection	SI-4 System Monitoring	supports	Detects anomalous integrity events or unexpected data changes
PBSAI-F-INT-002 Integrity Violation Detection	IR-4 Incident Handling	supports	Triggers investigation and response workflows for integrity violations
PBSAI-F-INT-003 Integrity Evidence Recording	AU-3 Content of Audit Records	strongly-supports	Records integrity status, actor, object, timestamp, and validation result
PBSAI-F-INT-003 Integrity Evidence Recording	AU-12 Audit Record Generation	equivalent	Produces structured evidence for integrity monitoring

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-INT-004 Integrity-Based Enforcement	AC-3 Access Enforcement	supports	Prevents use of data or artifacts that fail integrity validation
PBSAI-F-INT-004 Integrity-Based Enforcement	SI-10 Information Input Validation	supports	Treats failed integrity as invalid input condition
PBSAI-F-INT-005 Integrity Recovery Support	CP-10 System Recovery and Reconstitution	supports	Supports restoration from known-good data or backups after integrity failure

F-PRV — Privacy Governance and Purpose Limitation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-PRV-001 Purpose Limitation Enforcement	PT-2 Authority to Process Personally Identifiable Information	equivalent	Ensures PII processing is tied to authorized purpose
PBSAI-F-PRV-001 Purpose Limitation Enforcement	PT-3 Personally Identifiable Information Processing Purposes	equivalent	Enforces declared and approved processing purposes
PBSAI-F-PRV-002 Lawful Basis and Regulatory Validation	PT-2 Authority to Process Personally Identifiable Information	strongly-supports	Validates whether processing is authorized under applicable policy or law
PBSAI-F-PRV-002 Lawful Basis and Regulatory Validation	RA-3 Risk Assessment	supports	Evaluates privacy risk arising from data processing

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-PRV-003 Privacy Risk Identification and Escalation	PT-4 Consent	supports	Supports escalation when consent or processing authority is unclear
PBSAI-F-PRV-003 Privacy Risk Identification and Escalation	IR-4 Incident Handling	supports	Routes privacy-impacting events to review or response workflows
PBSAI-F-PRV-004 Privacy Attribute Propagation	AC-16 Security and Privacy Attributes	equivalent	Propagates privacy attributes into access, processing, and audit decisions
PBSAI-F-PRV-004 Privacy Attribute Propagation	AU-3 Content of Audit Records	supports	Preserves privacy context in audit records
PBSAI-F-PRV-005 Privacy Evidence Generation	AU-12 Audit Record Generation	supports	Records privacy-relevant decisions, processing context, and approvals
PBSAI-F-PRV-005 Privacy Evidence Generation	PT-7 Specific Categories of Personally Identifiable Information	supports	Supports accountability for sensitive categories of PII processing

F-DSP — Retention, Disposal, and Legal Hold

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-DSP-001 Retention Enforcement	SI-12 Information Management and Retention	equivalent	Enforces retention rules based on classification, policy, and regulation

PBSAI Control	NIST SP 800-53 Control	Relationship Rationale	
PBSAI-F-DSP-001 Retention Enforcement	AU-11 Audit Record Retention	supports	Supports retention of audit-relevant records
PBSAI-F-DSP-002 Secure and Verifiable Disposal	MP-6 Media Sanitization	equivalent	Ensures data disposal is controlled, verifiable, and policy-bound
PBSAI-F-DSP-002 Secure and Verifiable Disposal	AU-12 Audit Record Generation	supports	Produces disposal evidence and confirmation records
PBSAI-F-DSP-003 Legal-Hold Protection	AU-9 Protection of Audit Information	supports	Prevents modification or deletion of evidence under legal hold
PBSAI-F-DSP-003 Legal-Hold Protection	PM-30 Supply Chain Risk Management Strategy	supports	Supports enterprise-level governance of protected records and obligations
PBSAI-F-DSP-004 Disposal Authorization	AC-3 Access Enforcement	supports	Requires authorization before disposal actions execute
PBSAI-F-DSP-004 Disposal Authorization	AC-6 Least Privilege	supports	Restricts disposal authority to approved roles and processes
PBSAI-F-DSP-005 Retention and Disposal Evidence	AU-12 Audit Record Generation	equivalent	Produces structured evidence for retention and disposal decisions
PBSAI-F-DSP-005 Retention and Disposal Evidence	AU-6 Audit Record Review, Analysis, and Reporting	supports	Enables audit review of retention and disposal practices

F-EVID — Data Evidence, Lineage, and Handling Traceability

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-F-EVID-001 Data Action Evidence Records	AU-3 Content of Audit Records	strongly-supports	Captures data action context, actor, object, policy, and outcome
PBSAI-F-EVID-001 Data Action Evidence Records	AU-12 Audit Record Generation	equivalent	Requires data actions to generate structured audit records
PBSAI-F-EVID-002 Data Lineage and Handling Traceability	AU-12 Audit Record Generation	strongly-supports	Preserves traceability of data movement, transformation, and use
PBSAI-F-EVID-002 Data Lineage and Handling Traceability	AC-16 Security and Privacy Attributes	supports	Maintains classification and privacy metadata across handling events
PBSAI-F-EVID-003 Evidence Integrity for Data Operations	AU-9 Protection of Audit Information	equivalent	Protects evidence related to data operations from tampering
PBSAI-F-EVID-003 Evidence Integrity for Data Operations	AU-10 Non-Repudiation	strongly-supports	Supports attribution and accountability for data actions
PBSAI-F-EVID-004 Evidence Accessibility for Data Review	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Makes data-handling evidence available for audit, investigation, and privacy review
PBSAI-F-EVID-004 Evidence Accessibility for Data Review	CA-7 Continuous Monitoring	supports	Supports continuous assessment of data protection controls

Domain F — Cross-Domain Governance Alignment

Domain F provides the data-governance foundation for PBSAI's AI-enabled cybersecurity ecosystem.

It supplies:

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- data classification,
- privacy attributes,
- retention constraints,
- cryptographic posture,
- DLP events,
- integrity signals,
- lineage records,
- and evidence artifacts.

Domain F depends heavily on:

- Domain A for governance, risk, compliance, and policy authority,
- Domain B for asset and repository context,
- Domain C for identity and access context,
- Domain D for monitoring and anomaly detection,
- Domain E for enforcement and remediation,
- Domain G for incident handling and forensic preservation,
- Domain K for supply-chain and component provenance,
- and Domain L for analytics, data curation, and AI validation.

Domain F also operationalizes the following global PBSAI controls:

- PBSAI-GEN-MCP — context and policy binding,
- PBSAI-GEN-OC — structured output contracts,
- PBSAI-GEN-EVID — evidence generation,
- PBSAI-GEN-PROV — provenance and attribution,
- PBSAI-GEN-HITL — human authorization for high-impact data actions,
- PBSAI-GEN-SAFE — rollback and safe execution,
- PBSAI-GEN-FEED — continuous improvement.

Together, Domain F ensures that data security and privacy obligations are not treated as static policies. They become runtime-governed, evidence-producing, classification-aware, and privacy-context-aware operational controls.

Domain G — Incident Response and Forensics → NIST SP 800-53 Rev 5

Domain G defines PBSAI’s incident response, forensic analysis, evidence preservation, root-cause analysis, escalation, coordination, and post-incident learning capabilities.

Domain G operationalizes:

- governed incident handling,
- evidence-centric investigations,
- explainable response workflows,
- containment coordination,
- forensic integrity preservation,
- and continuous operational improvement.

The domain aligns strongly with:

- IR — Incident Response,
- AU — Audit and Accountability,
- SI — System and Information Integrity,
- CA — Security Assessment and Monitoring,
- CP — Contingency Planning,
- PM — Program Management,
- SC — System and Communications Protection,
- and RA — Risk Assessment.

G-IR — Incident Identification, Triage, and Coordination

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-IR-001 Incident Identification	IR-4 Incident Handling	equivalent	Identifies operational security incidents

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
			requiring governed response
PBSAI-G-IR-001 Incident Identification	SI-4 System Monitoring	strongly-supports	Uses telemetry and monitoring signals to detect incident conditions
PBSAI-G-IR-002 Incident Classification and Severity Assignment	IR-4 Incident Handling	strongly-supports	Categorizes incidents by severity, impact, and operational scope
PBSAI-G-IR-002 Incident Classification and Severity Assignment	RA-3 Risk Assessment	supports	Incorporates operational and business risk into classification
PBSAI-G-IR-003 Incident Triage and Routing	IR-5 Incident Monitoring	equivalent	Routes incidents to appropriate containment, response, and escalation workflows
PBSAI-G-IR-003 Incident Triage and Routing	AC-3 Access Enforcement	supports	Restricts access to incident workflows and operational actions
PBSAI-G-IR-004 Incident Escalation Management	IR-4 Incident Handling	equivalent	Escalates incidents based on severity, confidence, or authority constraints
PBSAI-G-IR-004 Incident Escalation Management	PBSAI-GEN-HITL-001 Human Approval Requirement	enhances	Requires human authorization for high-impact incident actions
PBSAI-G-IR-005 Incident Coordination and Workflow Tracking	IR-4 Incident Handling	strongly-supports	Coordinates multi-stage incident response activities

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-IR-005 Incident Coordination and Workflow Tracking	AU-12 Audit Record Generation	supports	Records workflow state transitions and operational decisions
PBSAI-G-IR-006 Incident Closure and Verification	IR-4 Incident Handling	strongly-supports	Validates resolution before formal incident closure
PBSAI-G-IR-006 Incident Closure and Verification	CA-7 Continuous Monitoring	supports	Confirms restored operational state after incident handling

G-FOR — Forensic Collection and Preservation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-FOR-001 Forensic Evidence Collection	AU-12 Audit Record Generation	equivalent	Collects structured evidence artifacts associated with incidents
PBSAI-G-FOR-001 Forensic Evidence Collection	IR-4 Incident Handling	strongly-supports	Supports investigation and incident reconstruction
PBSAI-G-FOR-002 Evidence Integrity Preservation	AU-9 Protection of Audit Information	equivalent	Protects forensic artifacts from unauthorized modification
PBSAI-G-FOR-002 Evidence Integrity Preservation	AU-10 Non-Repudiation	strongly-supports	Preserves evidentiary attribution and trustworthiness
PBSAI-G-FOR-003 Chain-of-Custody Enforcement	AU-10 Non-Repudiation	equivalent	Maintains attributable custody and handling records

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-FOR-003 Chain-of-Custody Enforcement	AU-3 Content of Audit Records	strongly-supports	Records handler, timestamp, transfer, and access details
PBSAI-G-FOR-004 Forensic Snapshot and Preservation	CP-9 System Backup	supports	Preserves recoverable operational state and forensic artifacts
PBSAI-G-FOR-004 Forensic Snapshot and Preservation	CP-10 System Recovery and Reconstitution	supports	Supports restoration and investigation after compromise
PBSAI-G-FOR-005 Forensic Access Control	AC-3 Access Enforcement	equivalent	Restricts forensic access to authorized investigators and systems
PBSAI-G-FOR-005 Forensic Access Control	AC-6 Least Privilege	strongly-supports	Limits evidence access scope and authority
PBSAI-G-FOR-006 Forensic Evidence Retention	AU-11 Audit Record Retention	equivalent	Retains evidence according to policy, legal, and operational requirements
PBSAI-G-FOR-006 Forensic Evidence Retention	MP-6 Media Sanitization	supports	Governs eventual disposal of retained evidence

G-RCA — Root Cause Analysis and Lessons Learned

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-RCA-001 Root Cause Identification	IR-4 Incident Handling	strongly-supports	Identifies technical and operational causes of incidents

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-RCA-001 Root Cause Identification	RA-3 Risk Assessment	supports	Evaluates systemic risk conditions contributing to incidents
PBSAI-G-RCA-002 Incident Correlation and Timeline Reconstruction	AU-6 Audit Record Review, Analysis, and Reporting	equivalent	Reconstructs incident sequences using evidence and telemetry
PBSAI-G-RCA-002 Incident Correlation and Timeline Reconstruction	AU-12 Audit Record Generation	strongly-supports	Correlates structured audit artifacts and execution records
PBSAI-G-RCA-003 Control Failure Analysis	CA-2 Control Assessments	strongly-supports	Evaluates why controls failed or were bypassed
PBSAI-G-RCA-003 Control Failure Analysis	SI-4 System Monitoring	supports	Uses operational telemetry to evaluate control performance
PBSAI-G-RCA-004 Lessons Learned and Feedback Integration	PM-14 Testing, Training, and Monitoring	equivalent	Uses incident outcomes to improve governance and operations
PBSAI-G-RCA-004 Lessons Learned and Feedback Integration	CA-7 Continuous Monitoring	strongly-supports	Feeds incident findings into monitoring and improvement workflows
PBSAI-G-RCA-005 Detection and Response Improvement	SI-4 System Monitoring	strongly-supports	Improves future detection and monitoring effectiveness
PBSAI-G-RCA-005 Detection and Response Improvement	IR-5 Incident Monitoring	supports	Refines response workflows and escalation logic
PBSAI-G-RCA-006 Governance and Policy Adjustment	PL-1 Policy and Procedures	supports	Updates governance procedures based on operational findings

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-RCA-006 Governance and Policy Adjustment	PM-9 Risk Management Strategy	supports	Adjusts enterprise risk posture and governance priorities

G-CONT — Containment and Operational Isolation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-CONT-001 Incident Containment Execution	IR-4 Incident Handling	equivalent	Executes containment actions to limit operational impact
PBSAI-G-CONT-001 Incident Containment Execution	SC-7 Boundary Protection	strongly-supports	Restricts movement and communication during containment
PBSAI-G-CONT-002 Dynamic Isolation and Segmentation	SC-32 System Partitioning	equivalent	Isolates affected systems, workloads, or operational domains
PBSAI-G-CONT-002 Dynamic Isolation and Segmentation	AC-4 Information Flow Enforcement	strongly-supports	Restricts flows between compromised and trusted environments
PBSAI-G-CONT-003 Containment Scope Governance	AC-6 Least Privilege	supports	Restricts containment authority to approved operational boundaries
PBSAI-G-CONT-003 Containment Scope Governance	CM-3 Configuration Change Control	supports	Governs operational changes introduced during containment
PBSAI-G-CONT-004 Emergency Containment Override	IR-4 Incident Handling	strongly-supports	Allows emergency response for critical incidents

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-CONT-004 Emergency Containment Override	PBSAI-GEN-SAFE-003 Kill Switch / Emergency Stop	enhances	Supports rapid operational interruption during unsafe conditions
PBSAI-G-CONT-005 Containment Verification	CA-7 Continuous Monitoring	strongly-supports	Verifies containment effectiveness and residual exposure
PBSAI-G-CONT-005 Containment Verification	SI-4 System Monitoring	supports	Monitors post-containment operational state

G-EVID — Incident Evidence and Traceability

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-EVID-001 Incident Evidence Generation	AU-2 Event Logging	equivalent	Requires incidents and response activities to generate evidence
PBSAI-G-EVID-001 Incident Evidence Generation	AU-12 Audit Record Generation	equivalent	Produces structured incident and response evidence artifacts
PBSAI-G-EVID-002 Incident Timeline Traceability	AU-3 Content of Audit Records	strongly-supports	Captures temporal sequence and operational context
PBSAI-G-EVID-002 Incident Timeline Traceability	AU-6 Audit Record Review, Analysis, and Reporting	supports	Enables investigation and incident reconstruction
PBSAI-G-EVID-003 Response Attribution and Accountability	AU-10 Non-Repudiation	equivalent	Attributes incident decisions and actions to authorized actors

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-G-EVID-003 Response Attribution and Accountability	IA-2 Identification and Authentication	supports	Links operational identity to incident actions
PBSAI-G-EVID-004 Cross-Domain Evidence Correlation	AU-12 Audit Record Generation	strongly-supports	Correlates telemetry, response, identity, and governance records
PBSAI-G-EVID-004 Cross-Domain Evidence Correlation	CA-7 Continuous Monitoring	supports	Supports continuous operational situational awareness
PBSAI-G-EVID-005 Evidence Accessibility and Review	AU-6 Audit Record Review, Analysis, and Reporting	equivalent	Makes incident evidence available for audit, compliance, and investigation
PBSAI-G-EVID-005 Evidence Accessibility and Review	CA-2 Control Assessments	supports	Supports evidence-based assessment and validation

Domain G — Cross-Domain Governance Alignment

Domain G acts as PBSAI's operational investigation and incident-governance layer.

The domain:

- coordinates response activities,
- preserves evidence integrity,
- supports forensic investigation,
- constrains containment execution,
- and feeds operational findings back into governance and detection systems.

Domain G relies heavily upon:

- Domain A for policy, governance, compliance, and risk guidance,
- Domain C for identity attribution and authorization,

- Domain D for telemetry, detections, and threat intelligence,
- Domain E for containment, enforcement, and remediation,
- Domain F for data-governance and evidence handling,
- Domain H for resilience and recovery workflows,
- and Domain L for analytics and feedback integration.

The domain operationalizes several global PBSAI governance controls:

- PBSAI-GEN-MCP — context and policy binding,
- PBSAI-GEN-EVID — evidence generation,
- PBSAI-GEN-PROV — provenance and attribution,
- PBSAI-GEN-HITL — human authorization,
- PBSAI-GEN-SAFE — safe execution and rollback,
- and PBSAI-GEN-FEED — operational feedback and continuous improvement.

Together, Domain G enables:

- explainable incident handling,
- evidence-centric investigation,
- bounded autonomous response,
- forensic accountability,
- and continuously improving defensive cybersecurity operations within AI-enabled enterprise environments.

Domain H — Resilience, Continuity, and Recovery → NIST SP 800-53 Rev 5

Domain H defines PBSAI's operational resilience, continuity assurance, backup governance, recovery orchestration, failover coordination, degradation management, and post-disruption restoration capabilities.

Domain H operationalizes:

- resilient operational behavior,
- continuity-aware governance,
- recovery verification,
- bounded restoration workflows,

- and fail-safe operational state management.

The domain aligns strongly with:

- CP — Contingency Planning,
- SI — System and Information Integrity,
- CA — Security Assessment and Monitoring,
- IR — Incident Response,
- AU — Audit and Accountability,
- SC — System and Communications Protection,
- PM — Program Management,
- and RA — Risk Assessment.

H-RES — Resilience and Continuity Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-RES-001 Operational Resilience Planning	CP-2 Contingency Plan	equivalent	Defines resilience and continuity procedures for operational disruption scenarios
PBSAI-H-RES-001 Operational Resilience Planning	PM-8 Critical Infrastructure Plan	strongly-supports	Aligns continuity planning with critical operational infrastructure
PBSAI-H-RES-002 Critical Function Identification	CP-2 Contingency Plan	strongly-supports	Identifies systems and functions requiring continuity prioritization
PBSAI-H-RES-002 Critical Function Identification	PM-11 Mission and Business Process Definition	supports	Maps operational continuity to mission-critical business functions
PBSAI-H-RES-003 Failure Isolation and Containment	SC-7 Boundary Protection	equivalent	Prevents cascading failure across interconnected systems and domains

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-RES-003 Failure Isolation and Containment	SC-24 Fail in Known State	strongly-supports	Ensures systems degrade safely under failure conditions
PBSAI-H-RES-004 Graceful Degradation Management	SI-13 Predictable Failure Prevention	equivalent	Maintains bounded operational behavior during degraded conditions
PBSAI-H-RES-004 Graceful Degradation Management	CP-10 System Recovery and Reconstitution	supports	Preserves recoverability while operating in degraded state
PBSAI-H-RES-005 Continuity Governance Validation	CA-2 Control Assessments	strongly-supports	Evaluates effectiveness of resilience and continuity controls
PBSAI-H-RES-005 Continuity Governance Validation	CA-7 Continuous Monitoring	supports	Monitors resilience posture and continuity readiness

H-BKP — Backup and Recovery Assurance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-BKP-001 Backup Definition and Governance	CP-9 System Backup	equivalent	Defines backup requirements, schedules, and governance constraints
PBSAI-H-BKP-001 Backup Definition and Governance	CM-9 Configuration Management Plan	supports	Governs operational backup configuration state

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-BKP-002 Backup Integrity Validation	SI-7 Software, Firmware, and Information Integrity	strongly-supports	Verifies integrity and trustworthiness of backup artifacts
PBSAI-H-BKP-002 Backup Integrity Validation	AU-9 Protection of Audit Information	supports	Protects backup evidence and integrity records
PBSAI-H-BKP-003 Backup Availability Verification	CP-9 System Backup	strongly-supports	Confirms recoverability and accessibility of backup assets
PBSAI-H-BKP-003 Backup Availability Verification	CA-7 Continuous Monitoring	supports	Continuously evaluates backup operational status
PBSAI-H-BKP-004 Backup Access Control	AC-3 Access Enforcement	equivalent	Restricts access to backup repositories and recovery operations
PBSAI-H-BKP-004 Backup Access Control	AC-6 Least Privilege	strongly-supports	Limits backup and recovery authority to approved roles
PBSAI-H-BKP-005 Backup Evidence and Auditability	AU-12 Audit Record Generation	equivalent	Produces evidence for backup creation, validation, and restoration
PBSAI-H-BKP-005 Backup Evidence and Auditability	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports audit and compliance review of backup operations

H-REC — Recovery and Restoration Operations

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-REC-001 Recovery Planning	CP-10 System Recovery and Reconstitution	equivalent	Defines procedures for restoration after disruption or compromise
PBSAI-H-REC-001 Recovery Planning	IR-4 Incident Handling	supports	Aligns recovery activities with incident response workflows
PBSAI-H-REC-002 Recovery Prioritization	RA-7 Risk Response	strongly-supports	Prioritizes restoration activities based on operational impact and risk
PBSAI-H-REC-002 Recovery Prioritization	PM-11 Mission and Business Process Definition	supports	Aligns recovery sequence with mission-critical operations
PBSAI-H-REC-003 Recovery Execution	CP-10 System Recovery and Reconstitution	equivalent	Executes controlled restoration and reconstitution operations
PBSAI-H-REC-003 Recovery Execution	CM-3 Configuration Change Control	strongly-supports	Governs operational changes during restoration
PBSAI-H-REC-004 Recovery Validation and Verification	CA-2 Control Assessments	strongly-supports	Confirms restored systems meet approved operational requirements
PBSAI-H-REC-004 Recovery Validation and Verification	SI-7 Software, Firmware, and Information Integrity	supports	Verifies restored integrity and trusted operational state
PBSAI-H-REC-005 Post-Recovery Monitoring	CA-7 Continuous Monitoring	equivalent	Monitors systems after restoration for residual instability or compromise
PBSAI-H-REC-005 Post-Recovery Monitoring	SI-4 System Monitoring	strongly-supports	Detects abnormal behavior following recovery

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-REC-006 Recovery Rollback and Safe Restoration	SC-24 Fail in Known State	supports	Ensures failed restoration attempts revert to safe operational conditions
PBSAI-H-REC-006 Recovery Rollback and Safe Restoration	CP-10 System Recovery and Reconstitution	strongly-supports	Supports controlled rollback during unsuccessful recovery

H-FAIL — Failover and Redundancy Management

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-FAIL-001 Failover Orchestration	CP-8 Telecommunications Services	supports	Coordinates operational failover between systems and services
PBSAI-H-FAIL-001 Failover Orchestration	CP-10 System Recovery and Reconstitution	strongly-supports	Enables continuity through alternate operational environments
PBSAI-H-FAIL-002 Redundancy Validation	CP-11 Alternate Communications Protocols	supports	Verifies availability of alternate operational capabilities
PBSAI-H-FAIL-002 Redundancy Validation	CA-7 Continuous Monitoring	supports	Continuously evaluates failover readiness
PBSAI-H-FAIL-003 Failover Scope Constraints	AC-6 Least Privilege	supports	Restricts failover authority and operational scope
PBSAI-H-FAIL-003 Failover Scope Constraints	SC-7 Boundary Protection	strongly-supports	Prevents failover operations from exceeding approved boundaries

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-FAIL-004 Automated Failover Governance	SI-13 Predictable Failure Prevention	strongly-supports	Enables deterministic failover behavior under operational stress
PBSAI-H-FAIL-004 Automated Failover Governance	PBSAI-GEN-SAFE-001 Safe Execution Constraint	enhances	Ensures failover remains bounded and policy-constrained
PBSAI-H-FAIL-005 Failover Evidence and Verification	AU-12 Audit Record Generation	equivalent	Produces evidence for failover initiation, execution, and restoration
PBSAI-H-FAIL-005 Failover Evidence and Verification	AU-3 Content of Audit Records	strongly-supports	Records operational context and failover outcomes

H-TEST — Resilience and Recovery Testing

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-TEST-001 Continuity and Recovery Testing	CP-4 Contingency Plan Testing	equivalent	Requires regular testing of continuity and recovery procedures
PBSAI-H-TEST-001 Continuity and Recovery Testing	CA-2 Control Assessments	strongly-supports	Evaluates operational effectiveness of resilience controls
PBSAI-H-TEST-002 Recovery Time and Objective Validation	PM-6 Measures of Performance	supports	Measures operational recovery effectiveness and timing objectives
PBSAI-H-TEST-002 Recovery Time and Objective Validation	CP-2 Contingency Plan	supports	Verifies continuity objectives are achievable

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-TEST-003 Recovery Scenario Simulation	IR-4 Incident Handling	supports	Exercises recovery coordination under realistic disruption scenarios
PBSAI-H-TEST-003 Recovery Scenario Simulation	SI-4 System Monitoring	supports	Evaluates operational monitoring effectiveness during disruption
PBSAI-H-TEST-004 Test Evidence Generation	AU-12 Audit Record Generation	equivalent	Produces evidence artifacts from resilience and recovery testing
PBSAI-H-TEST-004 Test Evidence Generation	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports governance review of continuity readiness
PBSAI-H-TEST-005 Improvement and Feedback Integration	PM-14 Testing, Training, and Monitoring	equivalent	Uses testing outcomes to improve resilience governance
PBSAI-H-TEST-005 Improvement and Feedback Integration	CA-7 Continuous Monitoring	strongly-supports	Integrates findings into continuous operational improvement

H-EVID — Resilience Evidence and Operational Traceability

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-EVID-001 Resilience Event Logging	AU-2 Event Logging	equivalent	Requires continuity and recovery actions to generate evidence events
PBSAI-H-EVID-001 Resilience Event Logging	AU-12 Audit Record Generation	equivalent	Produces structured operational evidence for resilience workflows

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-H-EVID-002 Recovery Traceability	AU-3 Content of Audit Records	strongly-supports	Records restoration actions, actors, timestamps, and outcomes
PBSAI-H-EVID-002 Recovery Traceability	AU-10 Non-Repudiation	supports	Preserves attribution and accountability for recovery actions
PBSAI-H-EVID-003 Cross-Domain Recovery Correlation	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Correlates recovery evidence across operational domains
PBSAI-H-EVID-003 Cross-Domain Recovery Correlation	CA-7 Continuous Monitoring	supports	Supports resilience situational awareness
PBSAI-H-EVID-004 Recovery Governance Accessibility	AU-6 Audit Record Review, Analysis, and Reporting	equivalent	Makes recovery evidence accessible for audit and governance review
PBSAI-H-EVID-004 Recovery Governance Accessibility	CA-2 Control Assessments	supports	Supports evidence-based continuity validation

Domain H — Cross-Domain Governance Alignment

Domain H functions as PBSAI's operational resilience and restoration layer.

The domain:

- maintains continuity during disruption,
- constrains restoration operations,
- governs failover behavior,
- validates recoverability,

- and ensures resilience workflows remain evidence-centric and policy-bound.

Domain H relies heavily upon:

- Domain A for governance, risk prioritization, and policy authority,
- Domain B for configuration and authoritative state management,
- Domain D for detection and disruption awareness,
- Domain E for enforcement and remediation coordination,
- Domain F for backup, data integrity, and retention constraints,
- Domain G for incident-driven recovery activation,
- and Domain L for monitoring, analytics, and operational improvement.

The domain operationalizes several global PBSAI governance controls:

- PBSAI-GEN-MCP — context and policy binding,
- PBSAI-GEN-EVID — evidence generation,
- PBSAI-GEN-PROV — provenance and attribution,
- PBSAI-GEN-SAFE — rollback, safe execution, and fail-safe behavior,
- PBSAI-GEN-HITL — human authorization for high-impact restoration actions,
- and PBSAI-GEN-FEED — resilience feedback and continuous improvement.

Together, Domain H enables:

- resilient AI-enabled enterprise operations,
- explainable restoration workflows,
- bounded recovery automation,
- evidence-centric continuity governance,
- and deterministic operational recovery under adverse conditions.

Domain I — Security Architecture and Systems Engineering → NIST SP 800-53 Rev 5

Domain I defines PBSAI's secure architecture governance, systems-security engineering, trust-boundary design, deployment validation, infrastructure hardening, and secure-by-design operational engineering capabilities.

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Domain I operationalizes:

- * security architecture governance,
- * secure systems engineering,
- * policy-bound infrastructure design,
- * trust-boundary enforcement,
- * architecture validation,
- * deployment assurance,
- * and secure lifecycle integration.

The domain aligns strongly with:

- * SA — System and Services Acquisition,
- * SC — System and Communications Protection,
- * CM — Configuration Management,
- * PL — Planning,
- * SI — System and Information Integrity,
- * RA — Risk Assessment,
- * SR — Supply Chain Risk Management,
- * and CA — Security Assessment and Monitoring.

I-ARCH — Security Architecture Governance

PBSAI Control Relationship	Rationale	NIST SP 800-53 Control
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PBSAI-I-ARCH-001 Security Architecture Definition Engineering Principles	equivalent	SA-8 Security and Privacy
design requirements		Defines security architecture principles and
PBSAI-I-ARCH-001 Security Architecture Definition Architectures	equivalent	PL-8 Security and Privacy
Establishes authoritative enterprise security architecture guidance		
PBSAI-I-ARCH-002 Architecture Review and Validation	CA-2 Control Assessments	
strongly-supports	Evaluates architectural conformance to governance and security objectives	
PBSAI-I-ARCH-002 Architecture Review and Validation	RA-3 Risk Assessment	
supports	Assesses architectural risk and systemic exposure	
PBSAI-I-ARCH-003 Trust Boundary Definition	SC-7 Boundary Protection	
equivalent	Establishes logical and operational trust boundaries	
PBSAI-I-ARCH-003 Trust Boundary Definition Enforcement	AC-4 Information Flow	
strongly-supports	Governs permitted flows between trust domains	
PBSAI-I-ARCH-004 Architecture Exception Governance Change Control	CM-3 Configuration	
supports	Governs exceptions to approved architectural standards	
PBSAI-I-ARCH-004 Architecture Exception Governance	RA-7 Risk Response	
supports	Requires explicit risk acceptance for architectural deviations	
PBSAI-I-ARCH-005 Architecture Traceability and Evidence Generation	AU-12 Audit Record	
strongly-supports	Produces evidence linking systems to approved architecture patterns	
PBSAI-I-ARCH-005 Architecture Traceability and Evidence Architectures	PL-8 Security and Privacy	
supports	Maintains traceable architecture governance artifacts	

I-SEG — Segmentation and Isolation Engineering

PBSAI Control Relationship	Rationale	NIST SP 800-53 Control
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PBSAI-I-SEG-001 Security Segmentation Design equivalent	Defines segmentation architecture for systems and operational domains	SC-7 Boundary Protection
PBSAI-I-SEG-001 Security Segmentation Design strongly-supports	Implements partitioned trust and operational isolation	SC-32 System Partitioning
PBSAI-I-SEG-002 Inter-Domain Communication Governance Enforcement equivalent	Restricts and governs communication between operational domains	AC-4 Information Flow Enforcement
PBSAI-I-SEG-002 Inter-Domain Communication Governance Protection strongly-supports	Prevents unauthorized cross-domain interaction	SC-7 Boundary Protection
PBSAI-I-SEG-003 Isolation Verification strongly-supports	Continuously validates segmentation and isolation effectiveness	CA-7 Continuous Monitoring
PBSAI-I-SEG-003 Isolation Verification supports	Detects unauthorized or anomalous inter-domain activity	SI-4 System Monitoring
PBSAI-I-SEG-004 Dynamic Segmentation Enforcement strongly-supports	Supports adaptive segmentation under changing operational conditions	SC-7 Boundary Protection
PBSAI-I-SEG-004 Dynamic Segmentation Enforcement Safe Execution Constraint enhances	Constrains dynamic segmentation actions within approved operational bounds	PBSAI-GEN-SAFE-001
PBSAI-I-SEG-005 Segmentation Auditability supports	Records segmentation changes and boundary-control operations	AU-3 Content of Audit Records

PBSAI-I-SEG-005 Segmentation Auditability	AU-12 Audit Record Generation
equivalent	Produces structured evidence for segmentation governance

I-INF — Infrastructure Hardening and Platform Engineering

PBSAI Control	NIST SP 800-53 Control	
Relationship	Rationale	
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PBSAI-I-INF-001 Infrastructure Hardening Baselines	CM-6 Configuration Settings	
equivalent	Defines approved secure baseline configurations	
PBSAI-I-INF-001 Infrastructure Hardening Baselines	CM-7 Least Functionality	
strongly-supports	Restricts unnecessary platform capabilities and services	
PBSAI-I-INF-002 Hardening Validation	CA-2 Control Assessments	
strongly-supports	Verifies infrastructure compliance with approved baselines	
PBSAI-I-INF-002 Hardening Validation	SI-7 Software, Firmware, and	
Information Integrity supports	Ensures hardened systems maintain trusted	
operational state		
PBSAI-I-INF-003 Unauthorized Configuration Detection	CM-3 Configuration Change	
Control	strongly-supports Detects unauthorized changes to infrastructure	
state		
PBSAI-I-INF-003 Unauthorized Configuration Detection	SI-4 System Monitoring	
supports	Monitors for operational drift and unauthorized modification	
PBSAI-I-INF-004 Secure Platform Provisioning	SA-10 Developer Configuration	
Management	equivalent	Governs secure provisioning of infrastructure and
platform resources		
PBSAI-I-INF-004 Secure Platform Provisioning	CM-2 Baseline Configuration	
strongly-supports	Ensures platforms inherit approved baseline configurations	

PBSAI-I-INF-005 Infrastructure Governance Evidence	AU-12 Audit Record Generation	equivalent	Produces evidence for hardening, validation, and provisioning operations
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PBSAI-I-INF-005 Infrastructure Governance Evidence	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports governance review of platform engineering activities
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I-DEP — Deployment and Release Security

PBSAI Control Relationship	Rationale	NIST SP 800-53 Control	
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PBSAI-I-DEP-001 Secure Deployment Governance	SA-10 Developer Configuration Management	equivalent	Governs secure deployment workflows and release processes
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PBSAI-I-DEP-001 Secure Deployment Governance	CM-3 Configuration Change Control	strongly-supports	Treats deployments as controlled operational changes
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PBSAI-I-DEP-002 Deployment Validation and Integrity	SI-7 Software, Firmware, and Information Integrity	equivalent	Validates integrity of deployment artifacts before release
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PBSAI-I-DEP-002 Deployment Validation and Integrity	SR-11 Component Authenticity	strongly-supports	Verifies authenticity and provenance of deployment components
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PBSAI-I-DEP-003 Deployment Scope Constraints	AC-6 Least Privilege	supports	Restricts deployment authority and operational scope
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PBSAI-I-DEP-003 Deployment Scope Constraints	CM-5 Access Restrictions for Change	strongly-supports	Limits deployment modification privileges
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| PBSAI-I-DEP-004 Rollback and Safe Release Management | CP-10 System Recovery and Reconstitution | supports | Supports restoration after failed or unsafe deployments |

| PBSAI-I-DEP-004 Rollback and Safe Release Management | PBSAI-GEN-SAFE-002 Rollback Requirement | enhances | Enforces governed rollback semantics for deployment operations |

| PBSAI-I-DEP-005 Deployment Evidence and Traceability | AU-12 Audit Record Generation | equivalent | Produces deployment audit trails and release evidence |

| PBSAI-I-DEP-005 Deployment Evidence and Traceability | AU-10 Non-Repudiation | strongly-supports | Preserves attributable deployment authorization and execution history |

I-SSE — Secure Systems Engineering Lifecycle

PBSAI Control	NIST SP 800-53 Control
Relationship	Rationale

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| PBSAI-I-SSE-001 Security Requirements Engineering | SA-8 Security and Privacy Engineering Principles | equivalent | Integrates security requirements into systems engineering processes |

| PBSAI-I-SSE-001 Security Requirements Engineering | PL-2 System Security and Privacy Plans | strongly-supports | Links security requirements to system planning activities |

| PBSAI-I-SSE-002 Threat Modeling and Design Analysis | RA-3 Risk Assessment | equivalent | Evaluates architectural threats and attack paths during design |

| PBSAI-I-SSE-002 Threat Modeling and Design Analysis | SA-15 Development Process, Standards, and Tools | supports | Integrates security analysis into engineering workflows |

PBSAI-I-SSE-003 Secure Design Validation	CA-2 Control Assessments
strongly-supports Verifies that engineered systems meet security design objectives	

PBSAI-I-SSE-003 Secure Design Validation	SI-7 Software, Firmware, and Information Integrity supports	Ensures integrity-preserving design and implementation

PBSAI-I-SSE-004 Engineering Change Governance	CM-3 Configuration Change Control	equivalent	Governs modifications to engineered system designs

PBSAI-I-SSE-004 Engineering Change Governance for Change	supports	CM-5 Access Restrictions for Change	Restricts authority to modify engineering baselines

PBSAI-I-SSE-005 Engineering Evidence and Traceability	AU-12 Audit Record Generation	strongly-supports Produces traceable engineering and validation evidence

PBSAI-I-SSE-005 Engineering Evidence and Traceability	PL-8 Security and Privacy Architectures	supports	Preserves traceability between architecture and implementation

I-ASSUR — Architectural Assurance and Verification

PBSAI Control Relationship	Rationale	NIST SP 800-53 Control	

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PBSAI-I-ASSUR-001 Architectural Assurance Validation	CA-2 Control Assessments
equivalent Validates whether systems satisfy architecture and security objectives	

PBSAI-I-ASSUR-001 Architectural Assurance Validation	SA-8 Security and Privacy Engineering Principles	strongly-supports Confirms adherence to secure engineering principles

PBSAI-I-ASSUR-002 Cross-Domain Dependency Validation	RA-3 Risk Assessment
supports	Evaluates systemic risk introduced by architectural dependencies
PBSAI-I-ASSUR-002 Cross-Domain Dependency Validation	SC-7 Boundary Protection
supports	Ensures dependency paths do not violate trust boundaries
PBSAI-I-ASSUR-003 Continuous Architectural Monitoring	CA-7 Continuous Monitoring
equivalent	Continuously evaluates architecture conformance and drift
PBSAI-I-ASSUR-003 Continuous Architectural Monitoring	SI-4 System Monitoring
strongly-supports	Detects architecture-related operational anomalies
PBSAI-I-ASSUR-004 Assurance Evidence Preservation	AU-9 Protection of Audit Information
strongly-supports	Protects architectural assurance evidence and validation records
PBSAI-I-ASSUR-004 Assurance Evidence Preservation	AU-12 Audit Record Generation
equivalent	Produces structured architectural assurance evidence
PBSAI-I-ASSUR-005 Governance Reporting and Traceability	AU-6 Audit Record Review, Analysis, and Reporting
strongly-supports	Supports architecture governance and audit review
PBSAI-I-ASSUR-005 Governance Reporting and Traceability	PM-6 Measures of Performance
supports	Enables architecture governance metrics and reporting

Domain I — Cross-Domain Governance Alignment

Domain I functions as PBSAI's secure-by-design engineering and architecture-governance layer.

The domain:

- * governs trust boundaries,
- * constrains infrastructure and deployment behavior,
- * validates architecture conformance,
- * operationalizes systems-security engineering,
- * and ensures deployments remain explainable, attributable, and policy-bound.

Domain I relies heavily upon:

- * Domain A for governance policy and risk authority,
- * Domain B for configuration and authoritative-state management,
- * Domain C for privileged engineering and deployment authorization,
- * Domain D for threat intelligence and architecture-driven detection requirements,
- * Domain E for enforcement and remediation integration,
- * Domain F for data-security and cryptographic engineering constraints,
- * Domain H for resilience-aware engineering design,
- * Domain K for supply-chain integrity and component provenance,
- * and Domain L for analytics, orchestration, and continuous-assurance feedback.

The domain operationalizes several global PBSAI governance controls:

- * PBSAI-GEN-MCP — context and policy binding,
- * PBSAI-GEN-EVID — evidence generation,
- * PBSAI-GEN-PROV — provenance and attestation,
- * PBSAI-GEN-SAFE — rollback and fail-safe execution,
- * PBSAI-GEN-HITL — human authorization for high-impact engineering actions,
- * and PBSAI-GEN-FEED — continuous architecture and assurance improvement.

Together, Domain I enables:

- * explainable secure systems engineering,
- * policy-bound deployment governance,
- * evidence-centric architecture assurance,
- * bounded operational engineering automation,
- * and resilient secure-by-design enterprise AI infrastructure.

Domain J — Physical and Environmental Security → NIST SP 800-53 Rev 5

Domain J defines PBSAI's physical security governance, facility monitoring, environmental protection, operational safety, OT/ICS-aware monitoring, physical access assurance, and environmental resilience capabilities.

Domain J operationalizes:

- physical access governance,
- environmental telemetry monitoring,
- operational facility assurance,
- physical-to-cyber correlation,
- safety-condition escalation,
- and evidence-centric facility security workflows.

The domain aligns strongly with:

- PE — Physical and Environmental Protection,
- AU — Audit and Accountability,
- IR — Incident Response,
- SI — System and Information Integrity,
- CP — Contingency Planning,
- SC — System and Communications Protection,
- and CA — Security Assessment and Monitoring.

J-PHY — Physical Access and Facility Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-PHY-001 Physical Access Governance	PE-2 Physical Access Authorizations	equivalent	Defines authorized access to facilities and protected operational areas
PBSAI-J-PHY-001 Physical Access Governance	AC-3 Access Enforcement	strongly-supports	Enforces authorized physical access decisions
PBSAI-J-PHY-002 Physical Access Monitoring	PE-6 Monitoring Physical Access	equivalent	Continuously monitors entry, exit, and facility-access events
PBSAI-J-PHY-002 Physical Access Monitoring	AU-2 Event Logging	supports	Produces physical-access event records
PBSAI-J-PHY-003 Physical Access Escalation	IR-4 Incident Handling	strongly-supports	Escalates suspicious or unauthorized physical-access events
PBSAI-J-PHY-003 Physical Access Escalation	AC-6 Least Privilege	supports	Restricts access escalation authority
PBSAI-J-PHY-004 Visitor and Temporary Access Governance	PE-7 Visitor Control	equivalent	Governs temporary facility access and visitor management
PBSAI-J-PHY-004 Visitor and Temporary Access Governance	AU-12 Audit Record Generation	supports	Produces traceable visitor-access records
PBSAI-J-PHY-005 Physical Access Evidence Preservation	AU-9 Protection of Audit Information	strongly-supports	Protects physical-access logs and evidentiary records

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-PHY-005 Physical Access Evidence Preservation	AU-10 Non- Repudiation	supports	Preserves attributable physical-access evidence

J-ENV — Environmental Monitoring and Safety

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-ENV-001 Environmental Telemetry Monitoring	PE-14 Temperature and Humidity Controls	equivalent	Monitors environmental conditions affecting operational safety and infrastructure
PBSAI-J-ENV-001 Environmental Telemetry Monitoring	SI-4 System Monitoring	strongly- supports	Integrates environmental telemetry into operational monitoring
PBSAI-J-ENV-002 Environmental Threshold Validation	PE-13 Fire Protection	supports	Detects environmental conditions associated with fire or facility hazards
PBSAI-J-ENV-002 Environmental Threshold Validation	SI-13 Predictable Failure Prevention	strongly- supports	Prevents unsafe operation under hazardous environmental conditions
PBSAI-J-ENV-003 Environmental Incident Escalation	IR-4 Incident Handling	equivalent	Escalates environmental anomalies requiring operational intervention
PBSAI-J-ENV-003 Environmental Incident Escalation	CP-2 Contingency Plan	supports	Activates continuity workflows during environmental disruption
PBSAI-J-ENV-004 Environmental Protection Enforcement	PE-15 Water Damage Protection	supports	Enforces environmental- protection mechanisms for critical systems

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-ENV-004 Environmental Protection Enforcement	PE-13 Fire Protection	supports	Integrates automated environmental safeguards
PBSAI-J-ENV-005 Environmental Evidence and Traceability	AU-12 Audit Record Generation	equivalent	Produces evidence for environmental events and safety actions
PBSAI-J-ENV-005 Environmental Evidence and Traceability	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Enables governance review of environmental conditions and incidents

J-OT — Operational Technology and Facility Systems Security

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-OT-001 OT/ICS Monitoring	SI-4 System Monitoring	equivalent	Monitors operational technology and industrial-control-system environments
PBSAI-J-OT-001 OT/ICS Monitoring	SC-7 Boundary Protection	strongly-supports	Protects OT environments from unauthorized interaction
PBSAI-J-OT-002 OT Segmentation and Isolation	SC-32 System Partitioning	equivalent	Segregates OT and enterprise operational environments
PBSAI-J-OT-002 OT Segmentation and Isolation	AC-4 Information Flow Enforcement	strongly-supports	Restricts communication between OT and IT domains

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-OT-003 OT Change and Configuration Governance	CM-3 Configuration Change Control	strongly-supports	Governs modifications to OT infrastructure and operational systems
PBSAI-J-OT-003 OT Change and Configuration Governance	CM-6 Configuration Settings	supports	Maintains secure OT configuration baselines
PBSAI-J-OT-004 OT Incident Escalation and Coordination	IR-4 Incident Handling	equivalent	Coordinates OT-related incident response and escalation
PBSAI-J-OT-004 OT Incident Escalation and Coordination	CP-10 System Recovery and Reconstitution	supports	Supports OT operational restoration after disruption
PBSAI-J-OT-005 OT Evidence and Operational Traceability	AU-12 Audit Record Generation	equivalent	Produces traceable OT operational and security evidence
PBSAI-J-OT-005 OT Evidence and Operational Traceability	AU-3 Content of Audit Records	strongly-supports	Captures OT operational context and event details

J-SAFE — Facility Safety and Emergency Operations

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-SAFE-001 Safety Event Detection	PE-13 Fire Protection	supports	Detects facility hazards and unsafe operational conditions

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-SAFE-001 Safety Event Detection	SI-13 Predictable Failure Prevention	strongly-supports	Prevents unsafe continuation of operations
PBSAI-J-SAFE-002 Emergency Shutdown and Isolation	SC-24 Fail in Known State	equivalent	Places systems into safe operational state during emergencies
PBSAI-J-SAFE-002 Emergency Shutdown and Isolation	PBSAI-GEN-SAFE-003 Kill Switch / Emergency Stop	enhances	Enables rapid interruption of unsafe operational conditions
PBSAI-J-SAFE-003 Safety Workflow Coordination	IR-4 Incident Handling	strongly-supports	Coordinates safety-response workflows and escalation
PBSAI-J-SAFE-003 Safety Workflow Coordination	CP-2 Contingency Plan	supports	Integrates facility emergencies into continuity planning
PBSAI-J-SAFE-004 Facility Recovery Coordination	CP-10 System Recovery and Reconstitution	equivalent	Coordinates restoration of safe facility operation
PBSAI-J-SAFE-004 Facility Recovery Coordination	CA-7 Continuous Monitoring	supports	Monitors facility state during restoration
PBSAI-J-SAFE-005 Safety Evidence Preservation	AU-12 Audit Record Generation	equivalent	Produces evidence for safety incidents and emergency actions
PBSAI-J-SAFE-005 Safety Evidence Preservation	AU-9 Protection of Audit Information	strongly-supports	Protects safety and emergency evidence artifacts

J-SURV — Surveillance and Situational Awareness

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-SURV-001 Surveillance Monitoring	PE-6 Monitoring Physical Access	strongly-supports	Uses surveillance telemetry to monitor protected operational areas
PBSAI-J-SURV-001 Surveillance Monitoring	SI-4 System Monitoring	supports	Integrates surveillance events into operational monitoring workflows
PBSAI-J-SURV-002 Behavioral and Movement Analysis	PE-3 Physical Access Control	supports	Detects anomalous or unauthorized movement patterns
PBSAI-J-SURV-002 Behavioral and Movement Analysis	IR-5 Incident Monitoring	strongly-supports	Supports real-time situational awareness and escalation
PBSAI-J-SURV-003 Cross-Domain Event Correlation	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Correlates surveillance, identity, environmental, and OT events
PBSAI-J-SURV-003 Cross-Domain Event Correlation	CA-7 Continuous Monitoring	supports	Enhances operational situational awareness
PBSAI-J-SURV-004 Surveillance Governance and Retention	AU-11 Audit Record Retention	equivalent	Governs surveillance-record retention and accessibility
PBSAI-J-SURV-004 Surveillance Governance and Retention	PT-2 Authority to Process Personally Identifiable Information	supports	Governs lawful handling of surveillance-associated PII
PBSAI-J-SURV-005 Surveillance Evidence Protection	AU-9 Protection of Audit Information	equivalent	Protects surveillance evidence from tampering or unauthorized access

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-SURV-005 Surveillance Evidence Protection	AU-10 Non-Repudiation	strongly-supports	Preserves attributable surveillance evidence

J-EVID — Physical and Environmental Evidence Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-EVID-001 Physical Security Event Logging	AU-2 Event Logging	equivalent	Requires physical-security and environmental events to generate evidence
PBSAI-J-EVID-001 Physical Security Event Logging	AU-12 Audit Record Generation	equivalent	Produces structured facility-security evidence artifacts
PBSAI-J-EVID-002 Environmental and OT Traceability	AU-3 Content of Audit Records	strongly-supports	Captures operational context for facility and OT events
PBSAI-J-EVID-002 Environmental and OT Traceability	AU-6 Audit Record Review, Analysis, and Reporting	supports	Supports operational investigation and review
PBSAI-J-EVID-003 Cross-Domain Correlation	AU-12 Audit Record Generation	strongly-supports	Correlates physical, environmental, cyber, and identity evidence
PBSAI-J-EVID-003 Cross-Domain Correlation	CA-7 Continuous Monitoring	supports	Supports integrated operational awareness
PBSAI-J-EVID-004 Evidence Accessibility and Governance	AU-6 Audit Record Review, Analysis, and Reporting	equivalent	Makes facility-security evidence available for governance review

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-J-EVID-004 Evidence Accessibility and Governance	CA-2 Control Assessments	supports	Supports evidence-based physical-security validation

Domain J — Cross-Domain Governance Alignment

Domain J functions as PBSAI’s physical-security and environmental-governance layer.

The domain:

- governs physical access,
- monitors environmental conditions,
- protects operational facilities,
- coordinates safety-response workflows,
- and correlates physical and cyber operational evidence.

Domain J relies heavily upon:

- Domain A for governance policy, compliance, and risk prioritization,
- Domain B for asset and infrastructure context,
- Domain C for physical identity and authorization management,
- Domain D for monitoring and anomaly-detection integration,
- Domain E for enforcement and operational response,
- Domain G for incident escalation and investigation,
- Domain H for continuity and recovery coordination,
- and Domain L for analytics, telemetry aggregation, and governance reporting.

The domain operationalizes several global PBSAI governance controls:

- PBSAI-GEN-MCP — context and policy binding,
- PBSAI-GEN-EVID — evidence generation,
- PBSAI-GEN-PROV — provenance and attribution,

- PBSAI-GEN-SAFE — safe execution and emergency interruption,
- PBSAI-GEN-HITL — human authorization for high-impact facility actions,
- and PBSAI-GEN-FEED — operational feedback and continuous improvement.

Together, Domain J enables:

- explainable physical-security governance,
- evidence-centric environmental monitoring,
- bounded OT and facility-security operations,
- resilient safety coordination,
- and integrated cyber-physical enterprise protection for AI-enabled operational environments.

Domain K — Supply Chain and Lifecycle Security → NIST SP 800-53 Rev 5

Domain K defines PBSAI's supply-chain governance, procurement assurance, component provenance, lifecycle integrity, dependency risk management, software-bill-of-materials (SBOM) governance, vendor evaluation, and secure disposal capabilities.

Domain K operationalizes:

- supply-chain trust validation,
- provenance-aware procurement,
- component authenticity assurance,
- lifecycle governance,
- dependency monitoring,
- and evidence-centric supply-chain security workflows.

The domain aligns strongly with:

- SR — Supply Chain Risk Management,
- SA — System and Services Acquisition,
- SI — System and Information Integrity,
- RA — Risk Assessment,
- CM — Configuration Management,
- AU — Audit and Accountability,

- MP — Media Protection,
- and CA — Security Assessment and Monitoring.

K-SCRM — Supply Chain Risk Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-SCRM-001 Supply Chain Risk Assessment	SR-2 Supply Chain Risk Management Plan	equivalent	Defines structured assessment of supplier and supply-chain risk
PBSAI-K-SCRM-001 Supply Chain Risk Assessment	RA-3 Risk Assessment	strongly-supports	Evaluates operational and systemic supply-chain exposure
PBSAI-K-SCRM-002 Supplier Trust Evaluation	SR-6 Supplier Assessments and Reviews	equivalent	Assesses supplier trustworthiness and governance posture
PBSAI-K-SCRM-002 Supplier Trust Evaluation	SA-9 External System Services	strongly-supports	Validates external service-provider security posture
PBSAI-K-SCRM-003 Supply Chain Risk Prioritization	RA-7 Risk Response	strongly-supports	Prioritizes mitigation based on supplier criticality and exposure
PBSAI-K-SCRM-003 Supply Chain Risk Prioritization	PM-9 Risk Management Strategy	supports	Aligns supply-chain prioritization with enterprise risk tolerance
PBSAI-K-SCRM-004 Continuous Supplier Monitoring	CA-7 Continuous Monitoring	equivalent	Continuously evaluates supplier operational and security posture
PBSAI-K-SCRM-004 Continuous Supplier Monitoring	SI-4 System Monitoring	strongly-supports	Detects emerging supplier-related operational risk

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-SCRM-005 Supply Chain Governance Evidence	AU-12 Audit Record Generation	equivalent	Produces traceable evidence for supplier evaluations and decisions
PBSAI-K-SCRM-005 Supply Chain Governance Evidence	AU-6 Audit Record Review, Analysis, and supports Reporting		Supports governance and audit review of supply-chain operations

K-PROC — Procurement and Vendor Assurance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-PROC-001 Secure Procurement Governance	SA-4 Acquisition Process	equivalent	Governs procurement using defined security and assurance requirements
PBSAI-K-PROC-001 Secure Procurement Governance	SR-3 Supply Chain Controls and Processes	strongly-supports	Integrates supply-chain controls into procurement workflows
PBSAI-K-PROC-002 Vendor Security Validation	SA-9 External System Services	equivalent	Validates security posture of external providers and vendors
PBSAI-K-PROC-002 Vendor Security Validation	SR-5 Acquisition Strategies, Tools, and Methods	supports	Applies structured evaluation methods to procurement decisions
PBSAI-K-PROC-003 Procurement Authorization and Approval	AC-3 Access Enforcement	supports	Requires authorization for procurement approval workflows
PBSAI-K-PROC-003 Procurement Authorization and Approval	AC-5 Separation of Duties	strongly-supports	Separates procurement recommendation and approval authority

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-PROC-004 Procurement Exception Management	RA-7 Risk Response	supports	Governs procurement exceptions and compensating controls
PBSAI-K-PROC-004 Procurement Exception Management	CM-3 Configuration Change Control	supports	Treats exception approvals as governed operational changes
PBSAI-K-PROC-005 Procurement Traceability and Evidence	AU-12 Audit Record Generation	equivalent	Produces evidence for procurement evaluations, approvals, and exceptions
PBSAI-K-PROC-005 Procurement Traceability and Evidence	AU-10 Non-Repudiation	strongly-supports	Preserves attributable procurement decision records

K-PROV — Provenance and Component Authenticity

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-PROV-001 Component Provenance Validation	SR-4 Provenance	equivalent	Validates origin and chain-of-custody for components and artifacts
PBSAI-K-PROV-001 Component Provenance Validation	SR-11 Component Authenticity	strongly-supports	Ensures components are authentic and unmodified
PBSAI-K-PROV-002 Cryptographic Provenance Verification	SI-7 Software, Firmware, and Information Integrity	equivalent	Uses cryptographic validation to verify integrity and provenance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-PROV-002 Cryptographic Provenance Verification	SC-12 Cryptographic Key Establishment and Management		Relies on secure key management for provenance assurance
PBSAI-K-PROV-003 Provenance Policy Enforcement	AC-3 Access Enforcement	supports	Prevents use of components lacking valid provenance
PBSAI-K-PROV-003 Provenance Policy Enforcement	SI-10 Information Input Validation	strongly-supports	Treats unverifiable provenance as invalid operational input
PBSAI-K-PROV-004 Provenance Drift Detection	SI-4 System Monitoring	strongly-supports	Detects changes in provenance status or authenticity posture
PBSAI-K-PROV-004 Provenance Drift Detection	CA-7 Continuous Monitoring	supports	Continuously evaluates provenance validity
PBSAI-K-PROV-005 Provenance Evidence Preservation	AU-9 Protection of Audit Information	equivalent	Protects provenance records and validation evidence
PBSAI-K-PROV-005 Provenance Evidence Preservation	AU-12 Audit Record Generation	strongly-supports	Produces traceable provenance validation records

K-SBOM — Software Bill of Materials and Dependency Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-SBOM-001 SBOM Generation and Maintenance	CM-8 System Component Inventory	equivalent	Maintains authoritative inventory of software components and dependencies

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-SBOM-001 SBOM Generation and Maintenance	SR-4 Provenance	strongly-supports	Preserves dependency provenance and lineage visibility
PBSAI-K-SBOM-002 Dependency Risk Analysis	RA-3 Risk Assessment	equivalent	Evaluates operational and security risk arising from dependencies
PBSAI-K-SBOM-002 Dependency Risk Analysis	RA-5 Vulnerability Monitoring and Scanning	strongly-supports	Detects vulnerable or high-risk dependencies
PBSAI-K-SBOM-003 Dependency Policy Enforcement	CM-7 Least Functionality	supports	Restricts use of unnecessary or prohibited components
PBSAI-K-SBOM-003 Dependency Policy Enforcement	AC-3 Access Enforcement	supports	Prevents deployment of unauthorized dependencies
PBSAI-K-SBOM-004 Continuous Dependency Monitoring	CA-7 Continuous Monitoring	equivalent	Continuously evaluates dependency security posture
PBSAI-K-SBOM-004 Continuous Dependency Monitoring	SI-4 System Monitoring	strongly-supports	Detects newly introduced dependency risk
PBSAI-K-SBOM-005 SBOM Evidence and Auditability	AU-12 Audit Record Generation	equivalent	Produces traceable dependency and SBOM evidence
PBSAI-K-SBOM-005 SBOM Evidence and Auditability	AU-6 Audit Record Review, Analysis, and Reporting	supports	Enables governance and audit review of dependency posture

K-LCM — Lifecycle Security and Secure Disposal

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PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-LCM-001 Secure Lifecycle Governance	SA-3 System Development Life Cycle	equivalent	Integrates security governance across system lifecycle phases
PBSAI-K-LCM-001 Secure Lifecycle Governance	SA-15 Development Process, Standards, and Tools	strongly-supports	Applies secure engineering practices throughout lifecycle activities
PBSAI-K-LCM-002 Lifecycle Change Governance	CM-3 Configuration Change Control	equivalent	Governs lifecycle modifications and operational transitions
PBSAI-K-LCM-002 Lifecycle Change Governance	CM-5 Access Restrictions for Change	supports	Restricts authority for lifecycle modifications
PBSAI-K-LCM-003 End-of-Life Risk Management	RA-7 Risk Response	strongly-supports	Evaluates and mitigates risk associated with unsupported systems/components
PBSAI-K-LCM-003 End-of-Life Risk Management	SI-2 Flaw Remediation	supports	Addresses security exposure from obsolete components
PBSAI-K-LCM-004 Secure Disposal and Sanitization	MP-6 Media Sanitization	equivalent	Ensures secure disposal of systems, media, and components
PBSAI-K-LCM-004 Secure Disposal and Sanitization	SR-12 Component Disposal	strongly-supports	Governs secure disposal of supply-chain-relevant components
PBSAI-K-LCM-005 Lifecycle Traceability and Evidence	AU-12 Audit Record Generation	equivalent	Produces lifecycle governance, disposal, and transition evidence

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-LCM-005 Lifecycle Traceability and Evidence	AU-10 Non- Repudiation	strongly- supports	Preserves attributable lifecycle decision records

K-ASSUR — Supply Chain Assurance and Validation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-ASSUR-001 Supply Chain Assurance Validation	CA-2 Control Assessments	equivalent	Validates effectiveness of supply-chain governance controls
PBSAI-K-ASSUR-001 Supply Chain Assurance Validation	SR-2 Supply Chain Risk Management Plan	strongly- supports	Confirms alignment with supply-chain governance requirements
PBSAI-K-ASSUR-002 Cross-Supplier Dependency Analysis	RA-3 Risk Assessment	supports	Evaluates systemic risk from supplier interdependencies
PBSAI-K-ASSUR-002 Cross-Supplier Dependency Analysis	PM-30 Supply Chain Risk Management Strategy	supports	Aligns dependency analysis with enterprise supply-chain strategy
PBSAI-K-ASSUR-003 Continuous Assurance Monitoring	CA-7 Continuous Monitoring	equivalent	Continuously evaluates supplier and dependency assurance posture
PBSAI-K-ASSUR-003 Continuous Assurance Monitoring	SI-4 System Monitoring	strongly- supports	Detects assurance degradation or emerging supplier anomalies
PBSAI-K-ASSUR-004 Assurance Evidence Protection	AU-9 Protection of Audit Information	equivalent	Protects assurance evidence and supplier validation artifacts

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-K-ASSUR-004 Assurance Evidence Protection	AU-12 Audit Record Generation	strongly-supports	Produces structured supply-chain assurance evidence
PBSAI-K-ASSUR-005 Governance Reporting and Traceability	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Supports supply-chain governance review and auditability
PBSAI-K-ASSUR-005 Governance Reporting and Traceability	PM-6 Measures of Performance	supports	Enables supply-chain governance metrics and reporting

Domain K — Cross-Domain Governance Alignment

Domain K functions as PBSAI's supply-chain trust and lifecycle-governance layer.

The domain:

- validates supplier trustworthiness,
- governs procurement and component provenance,
- constrains dependency risk,
- operationalizes SBOM-aware security governance,
- and preserves evidence-centric lifecycle traceability.

Domain K relies heavily upon:

- Domain A for governance policy, compliance, and enterprise risk authority,
- Domain B for asset and configuration inventory context,
- Domain C for procurement and approval authorization,
- Domain D for supplier-threat intelligence and exposure monitoring,
- Domain E for enforcement and remediation integration,
- Domain F for data-protection and retention constraints,

- Domain I for secure engineering and deployment validation,
- and Domain L for analytics, orchestration, and continuous-assurance reporting.

The domain operationalizes several global PBSAI governance controls:

- PBSAI-GEN-MCP — context and policy binding,
- PBSAI-GEN-EVID — evidence generation,
- PBSAI-GEN-PROV — provenance and cryptographic attestation,
- PBSAI-GEN-HITL — human authorization for high-impact procurement and lifecycle actions,
- PBSAI-GEN-SAFE — rollback and bounded operational execution,
- and PBSAI-GEN-FEED — supply-chain feedback and continuous improvement.

Together, Domain K enables:

- explainable supply-chain governance,
- provenance-aware procurement assurance,
- evidence-centric lifecycle security,
- bounded dependency and vendor-risk management,
- and resilient enterprise AI infrastructure trust validation.

Domain L — Security Program, Orchestration, Analytics, and AI Assurance → NIST SP 800-53 Rev 5

Domain L defines PBSAI's enterprise security-program orchestration, governance analytics, AI assurance, operational telemetry fusion, control-performance analysis, autonomous workflow coordination, and continuous-improvement capabilities.

Domain L operationalizes:

- enterprise security orchestration,
- governance analytics,
- AI-assisted assurance validation,
- operational intelligence fusion,
- program-level telemetry correlation,
- and evidence-centric continuous improvement.

The domain aligns strongly with:

- CA — Security Assessment and Monitoring,
- PM — Program Management,
- AU — Audit and Accountability,
- SI — System and Information Integrity,
- RA — Risk Assessment,
- SA — System and Services Acquisition,
- IR — Incident Response,
- and PL — Planning.

L-ORCH — Security Orchestration and Coordination

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-ORCH-001 Security Workflow Orchestration	CA-7 Continuous Monitoring	strongly-supports	Coordinates monitoring, enforcement, response, and governance workflows
PBSAI-L-ORCH-001 Security Workflow Orchestration	IR-4 Incident Handling	supports	Integrates orchestration with response execution workflows
PBSAI-L-ORCH-002 Cross-Domain Coordination	PM-1 Information Security Program Plan	equivalent	Coordinates enterprise security functions across operational domains
PBSAI-L-ORCH-002 Cross-Domain Coordination	CA-7 Continuous Monitoring	strongly-supports	Correlates monitoring and governance activity across the ecosystem
PBSAI-L-ORCH-003 Workflow Authorization Governance	AC-3 Access Enforcement	supports	Restricts orchestration authority to approved operational scope

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-ORCH-003 Workflow Authorization Governance	AC-5 Separation of Duties	strongly-supports	Separates orchestration recommendation and execution authority
PBSAI-L-ORCH-004 Autonomous Workflow Constraint Enforcement	SC-24 Fail in Known State	supports	Constrains orchestration behavior under unsafe operational conditions
PBSAI-L-ORCH-004 Autonomous Workflow Constraint Enforcement	PBSAI-GEN-SAFE-001 Safe Execution Constraint	enhances	Applies bounded operational semantics to orchestration workflows
PBSAI-L-ORCH-005 Orchestration Evidence and Traceability	AU-12 Audit Record Generation	equivalent	Produces traceable orchestration execution evidence
PBSAI-L-ORCH-005 Orchestration Evidence and Traceability	AU-10 Non-Repudiation	strongly-supports	Preserves attributable workflow-execution history

L-ANL — Governance Analytics and Operational Intelligence

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-ANL-001 Governance Metrics Collection	PM-6 Measures of Performance	equivalent	Defines and collects enterprise governance and operational metrics
PBSAI-L-ANL-001 Governance Metrics Collection	CA-7 Continuous Monitoring	strongly-supports	Supports operational-state awareness and continuous evaluation
PBSAI-L-ANL-002 Cross-Domain Telemetry Correlation	SI-4 System Monitoring	equivalent	Correlates telemetry across security,

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
			infrastructure, and governance domains
PBSAI-L-ANL-002 Cross-Domain Telemetry Correlation	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Enables integrated operational analysis and investigation
PBSAI-L-ANL-003 Risk and Exposure Analytics	RA-3 Risk Assessment	equivalent	Produces operational risk and exposure analysis
PBSAI-L-ANL-003 Risk and Exposure Analytics	RA-7 Risk Response	strongly-supports	Supports prioritization of mitigation and governance actions
PBSAI-L-ANL-004 Operational Trend and Drift Detection	SI-4 System Monitoring	strongly-supports	Detects operational anomalies and governance drift
PBSAI-L-ANL-004 Operational Trend and Drift Detection	CA-7 Continuous Monitoring	supports	Enables continuous operational-state validation
PBSAI-L-ANL-005 Analytics Evidence Preservation	AU-9 Protection of Audit Information	equivalent	Protects operational analytics evidence and derived findings
PBSAI-L-ANL-005 Analytics Evidence Preservation	AU-12 Audit Record Generation	strongly-supports	Produces structured analytics evidence artifacts

L-AI — AI Assurance and Model Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-AI-001 AI Model Governance	SA-8 Security and Privacy Engineering Principles	strongly-supports	Applies governed engineering principles to AI-enabled operational systems
PBSAI-L-AI-001 AI Model Governance	PM-9 Risk Management Strategy	supports	Aligns AI governance with enterprise risk-management objectives
PBSAI-L-AI-002 AI Output Validation	SI-10 Information Input Validation	equivalent	Validates AI-generated outputs before operational acceptance or execution
PBSAI-L-AI-002 AI Output Validation	AU-12 Audit Record Generation	supports	Produces evidence for AI-output validation decisions
PBSAI-L-AI-003 AI Confidence and Explainability Enforcement	AU-3 Content of Audit Records	strongly-supports	Preserves confidence scores, rationale, and explainability metadata
PBSAI-L-AI-003 AI Confidence and Explainability Enforcement	CA-2 Control Assessments	supports	Enables assessment of AI-governance effectiveness
PBSAI-L-AI-004 Human Oversight for High-Impact AI Decisions	AC-5 Separation of Duties	supports	Separates AI recommendation from human authorization
PBSAI-L-AI-004 Human Oversight for High-Impact AI Decisions	PBSAI-GEN-HITL-001 Human Approval Requirement	enhances	Requires human authorization for high-impact AI actions
PBSAI-L-AI-005 AI Drift and Behavior Monitoring	SI-4 System Monitoring	equivalent	Detects abnormal or drifting AI behavior

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-AI-005 AI Drift and Behavior Monitoring	CA-7 Continuous Monitoring	strongly-supports	Continuously evaluates AI operational trustworthiness
PBSAI-L-AI-006 AI Assurance Evidence and Traceability	AU-12 Audit Record Generation	equivalent	Produces evidence for AI decisions, validations, and governance workflows
PBSAI-L-AI-006 AI Assurance Evidence and Traceability	AU-10 Non-Repudiation	strongly-supports	Preserves attributable AI-governance evidence

L-PROG — Enterprise Security Program Governance

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-PROG-001 Security Program Governance	PM-1 Information Security Program Plan	equivalent	Defines enterprise-wide security governance structure and objectives
PBSAI-L-PROG-001 Security Program Governance	PL-1 Policy and Procedures	strongly-supports	Aligns program governance with operational security policies
PBSAI-L-PROG-002 Program Risk Integration	PM-9 Risk Management Strategy	equivalent	Integrates operational risk into program governance decisions
PBSAI-L-PROG-002 Program Risk Integration	RA-3 Risk Assessment	strongly-supports	Uses operational risk analysis to guide program priorities
PBSAI-L-PROG-003 Program Performance Monitoring	PM-6 Measures of Performance	equivalent	Evaluates security-program effectiveness and operational maturity

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-PROG-003 Program Performance Monitoring	CA-7 Continuous Monitoring	strongly-supports	Uses telemetry and evidence to assess program health
PBSAI-L-PROG-004 Governance Reporting and Escalation	AU-6 Audit Record Review, Analysis, and Reporting	equivalent	Produces governance and executive-level security reporting
PBSAI-L-PROG-004 Governance Reporting and Escalation	IR-6 Incident Reporting	supports	Escalates significant operational or governance events
PBSAI-L-PROG-005 Program Improvement and Adaptation	PM-14 Testing, Training, and Monitoring	equivalent	Continuously improves enterprise security governance processes
PBSAI-L-PROG-005 Program Improvement and Adaptation	CA-7 Continuous Monitoring	strongly-supports	Uses operational findings to improve governance effectiveness

L-ASSUR — Continuous Assurance and Validation

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-ASSUR-001 Continuous Assurance Evaluation	CA-7 Continuous Monitoring	equivalent	Continuously evaluates enterprise control effectiveness and assurance posture
PBSAI-L-ASSUR-001 Continuous Assurance Evaluation	CA-2 Control Assessments	strongly-supports	Supports ongoing validation of implemented controls
PBSAI-L-ASSUR-002 Cross-Domain Assurance Correlation	AU-6 Audit Record Review, Analysis, and Reporting	strongly-supports	Correlates evidence across operational and governance domains

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-ASSUR-002 Cross-Domain Assurance Correlation	SI-4 System Monitoring	supports	Integrates monitoring evidence into assurance workflows
PBSAI-L-ASSUR-003 Assurance Drift Detection	SI-4 System Monitoring	equivalent	Detects degradation in operational assurance posture
PBSAI-L-ASSUR-003 Assurance Drift Detection	RA-3 Risk Assessment	strongly-supports	Evaluates emerging operational risk from degraded assurance conditions
PBSAI-L-ASSUR-004 Assurance Escalation and Remediation Coordination	IR-4 Incident Handling	supports	Escalates critical assurance failures for remediation
PBSAI-L-ASSUR-004 Assurance Escalation and Remediation Coordination	RA-7 Risk Response	strongly-supports	Coordinates remediation of assurance deficiencies
PBSAI-L-ASSUR-005 Assurance Evidence Protection	AU-9 Protection of Audit Information	equivalent	Protects assurance evidence and validation records
PBSAI-L-ASSUR-005 Assurance Evidence Protection	AU-12 Audit Record Generation	strongly-supports	Produces traceable assurance and validation evidence

L-FEED — Feedback, Learning, and Continuous Improvement

PBSAI Control	NIST SP 800-53 Control	Relationship	Rationale
PBSAI-L-FEED-001 Operational Feedback Collection	PM-14 Testing, Training, and Monitoring	equivalent	Collects operational outcomes and governance findings for improvement
PBSAI-L-FEED-001 Operational Feedback Collection	CA-7 Continuous Monitoring	strongly-supports	Integrates operational evidence into improvement workflows
PBSAI-L-FEED-002 Cross-Domain Learning Integration	PM-1 Information Security Program Plan	supports	Feeds lessons learned across enterprise governance domains
PBSAI-L-FEED-002 Cross-Domain Learning Integration	IR-4 Incident Handling	supports	Incorporates incident-response outcomes into governance adaptation
PBSAI-L-FEED-003 Control Improvement Recommendation	RA-7 Risk Response	strongly-supports	Recommends governance and control improvements based on operational findings
PBSAI-L-FEED-003 Control Improvement Recommendation	PM-6 Measures of Performance	supports	Uses metrics to guide improvement priorities
PBSAI-L-FEED-004 Governance Adaptation and Optimization	PM-14 Testing, Training, and Monitoring	equivalent	Continuously adapts governance processes and operational workflows
PBSAI-L-FEED-004 Governance Adaptation and Optimization	CA-2 Control Assessments	supports	Validates effectiveness of implemented improvements
PBSAI-L-FEED-005 Improvement Evidence and Traceability	AU-12 Audit Record Generation	equivalent	Produces evidence for governance improvements and optimization actions
PBSAI-L-FEED-005 Improvement Evidence and Traceability	AU-10 Non-Repudiation	strongly-supports	Preserves attributable improvement and governance records

Domain L — Cross-Domain Governance Alignment

Domain L functions as PBSAI's enterprise governance, orchestration, analytics, and AI-assurance layer.

The domain:

- coordinates enterprise security workflows,
- aggregates operational telemetry,
- validates AI-assisted operational behavior,
- monitors governance effectiveness,
- and drives continuous assurance and improvement across all PBSAI domains.

Domain L depends heavily upon:

- Domain A for governance authority, policy, and enterprise risk strategy,
- Domains B through K for operational telemetry, evidence, and domain-specific control outputs,
- and the global PBSAI governance controls for context binding, provenance, evidence generation, authorization, safety, and continuous feedback.

The domain operationalizes several global PBSAI governance controls:

- PBSAI-GEN-MCP — context and policy binding,
- PBSAI-GEN-OC — output contract governance,
- PBSAI-GEN-EVID — evidence generation,
- PBSAI-GEN-PROV — provenance and attribution,
- PBSAI-GEN-HITL — human oversight and authorization,
- PBSAI-GEN-SAFE — bounded execution and rollback,
- and PBSAI-GEN-FEED — continuous learning and improvement.

Together, Domain L enables:

- explainable enterprise security orchestration,
- evidence-centric AI assurance,
- continuous operational governance validation,

- bounded autonomous security operations,
- and adaptive, resilient enterprise cybersecurity governance for AI-enabled environments.

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